

Quantifying Emergent Gravity in the Axis Model: The One-Loop Map to $G_{\text{eff}}(x)$

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Abstract

We quantify the emergent Newton coupling in the Axis Model. Starting from scalar-coherent projection-coarse-graining, we compute the projected one-loop effective action and obtain a closed-form, *local* expression for the Einstein–Hilbert coefficient in coherent domains,

$$s(x) \equiv \frac{G_{\text{eff}}(x)}{G_0} = \frac{1}{1 + \beta [1 - f(\Phi(x))]}, \quad \beta \equiv -\frac{G_0 \Lambda_\Phi^2}{2\pi} S, \quad S = \sum_s w_s \kappa_s.$$

We prove that this map is invariant under trivial field reparametrizations that preserve the coherence filter and is gauge/BRST-independent at one loop. We establish boundedness and continuity— $s \in [1/(1 + \beta), 1]$, $ds/df \geq 0$ —ensuring well-posed inference in the EFT window (weak curvature, slow variation of G_{eff}). In the weak-field, single-factor regime the result justifies the observational closure $G_{\text{eff}} = sG_0$ and yields concrete, falsifiable relations among observables (linear scaling of weak-lensing fields; predictable rescalings of Einstein angles, time-delay differences, and standard-siren amplitudes), including the joint invariant $R_{\Delta t} = (R_\theta)^2$. We also specify pre-registered tests and deterministic artifacts (figures, checksums, code hash) that fix the empirical surface for validation. Higher-order effects enter as $\mathcal{O}(R^2, \nabla R)$ and RG evolution of the EH coefficient; they do not remove the EH term within the stated domain. The paper thus places G_{eff} on a first-principles footing and connects it directly to data via a simple, testable parameter s .

1 Introduction

Gravity as we observe it may be an *emergent* macroscopic phenomenon rather than a fundamental interaction. Classic proposals (induced gravity, thermodynamic/holographic derivations) suggest that Einstein dynamics arise after coarse-graining over microscopic degrees of freedom, with the Einstein–Hilbert (EH) term appearing as the leading operator in a low-curvature expansion.[1, 2, 3, 4, 5] In this perspective, the central theoretical question is not whether an EH term appears—it does under broad conditions—but how its *coefficient* is fixed by the underlying microphysics and by the state of coherence of the relevant fields.

Axis Model approach. We adopt the Axis Model’s conditional coarse-graining over scalar-coherent microstates,

$$e^{iS_{\text{eff}}[g]} = \int_{\text{coherent}; g[v, \Phi]=g} \mathcal{D}v \mathcal{D}\Phi e^{iS[v, \Phi]}, \quad (1)$$

which yields, in the low-energy/low-curvature regime,

$$S_{\text{eff}}[g] = \frac{1}{16\pi G_{\text{eff}}(x)} \int d^4x \sqrt{-g} R + \mathcal{O}(R^2, \nabla R, \dots). \quad (2)$$

The scalar coherence filter $f(\Phi)$ projects onto domains where the internal fields (v_x, v_z, Φ) act coherently; within such domains, the projected one-loop effective action can be computed with

heat-kernel methods.¹ A key output is a *local* Newton coupling $G_{\text{eff}}(x)$ that depends on the coherent state, with a single-factor observable surrogate $s(x) \equiv G_{\text{eff}}(x)/G_0$ linking directly to lensing and GW observables.

What this paper solves. We provide a closed-form, one-loop expression for the EH coefficient in scalar-coherent domains, derive the corresponding map

$$s(x) = \frac{G_{\text{eff}}(x)}{G_0} = \frac{1}{1 + \beta [1 - f(x)]}, \quad \beta \equiv -\frac{G_0 \Lambda_\Phi^2}{2\pi} S, \quad S = \sum_s w_s \kappa_s,$$

and establish its *formal properties* (reparametrization and gauge/BRST invariances; boundedness and Lipschitz continuity). These results (i) quantify G_{eff} from first principles, and (ii) justify the single-factor observational closures used in Section 9—linking s to weak lensing, Einstein angles, strong-lensing time delays, and standard-siren amplitudes. On the empirical side, we point to pre-registered tests (Einstein angle, time delays, sirens) and a reproducible, deterministic notebook that implements the closed-form map for toy systems, along with checksums and environment capture.

Context and observables. The EH-as-leading-operator viewpoint is standard in effective-field-theory gravity,[3] and the heat-kernel/Seeley–DeWitt expansion provides the right bookkeeping for the curvature-linear coefficient and its higher-order corrections.[4, 5] On the data side, strong/weak lensing and time-delay cosmography supply clean, geometric observables with well-studied degeneracies (e.g., mass-sheet transformations),[6, 7, 8] while gravitational-wave *standard sirens* provide an amplitude-calibrated probe with independent systematics.[9, 10] Our single-factor map ties these sectors together in a way that is falsifiable by joint relations (e.g., $R_{\Delta t} = (R_\theta)^2$).

Roadmap. We first summarize the emergence setup and one-loop master identity; then we derive the matched form of $G_{\text{eff}}(x)$ and its formal properties (§6), establish the EFT power-counting/validity window and gradient bounds, and develop the observable closures and degeneracy audit (§9). We conclude with pre-registered tests and a reproducible artifact suite (App. F).

2 Field Content, Projection Geometry, and Backgrounds

2.1 Fields, projector, and emergent frame

Field content and indices. We consider two internal displacement fields $v_x(x)$ and $v_z(x)$ and a complex scalar $\Phi(x) = \rho(x) e^{i\theta(x)}$. Latin frame indices $a, b \in \{0, 1, 2, 3\}$ are raised/lowered with $\eta_{ab} = \text{diag}(+1, -1, -1, -1)$; spacetime indices $\mu, \nu \in \{0, 1, 2, 3\}$ are raised/lowered with $g_{\mu\nu}$ once it is defined. Prior to projection, $v_{x,z}$ are *internal* fields; their interpretation as spacetime vectors is conditional on scalar coherence.

Coherence scale and filter. Let Λ_Φ denote the scalar coherence scale. A region is *scalar-coherent* when $|\nabla_\mu \theta|^2 \ll \Lambda_\Phi^2$. We implement coherence as a local multiplicative filter

$$f(\Phi(x)) \equiv \exp \left[-\frac{|\nabla_\mu \theta(x)|^2}{\Lambda_\Phi^2} \right], \quad 0 < f(\Phi) \leq 1, \quad (3)$$

which smoothly suppresses projection as phase gradients grow (cf. Eq. (2) of [11]). In fully decoherent domains $f(\Phi) \rightarrow 0$, and geometric observables are undefined.

¹Projection geometry, the conditional integral, and the emergence of the EH lead term are derived in detail later in the manuscript; see also the heat-kernel references in §6.

Local projection map. The scalar-induced projection of an internal displacement into the spacetime tangent bundle is defined by

$$P_\Phi^\mu[v_a](x) = f(\Phi(x)) \lambda^\mu{}_a(x) v_a(x), \quad a \in \{x, z\}, \quad (4)$$

where $\lambda^\mu{}_a(x)$ is a non-dynamical alignment map fixed by the local scalar-phase gradient and the orientation of the alignment frame (cf. Eq. (9) and surrounding text in [11]). In coherent domains $\lambda^\mu{}_a$ varies slowly and furnishes spacelike legs of the emergent frame.

Emergent vierbein. The vierbein $e^a{}_\mu(x)$ is composite:

$$e^0{}_\mu(x) = \alpha_\Phi \partial_\mu \Phi(x), \quad e^1{}_\mu(x) = P_{\Phi\mu}[v_x](x), \quad e^2{}_\mu(x) = P_{\Phi\mu}[v_z](x), \quad (5)$$

with α_Φ a normalization factor of mass dimension $[E]^{-2}$ (cf. Eq. (10) of [11]). The fourth leg $e^3{}_\mu$ is obtained *without circularity* by Gram–Schmidt completion with respect to an auxiliary vector $n_\mu(x)$ linearly independent of $e^{0,1,2}{}_\mu$:

$$V_\mu(x) \equiv n_\mu(x) - (\eta^{\alpha\beta} n_\alpha e^0{}_\beta) e^0{}_\mu - (\eta^{\alpha\beta} n_\alpha e^1{}_\beta) e^1{}_\mu - (\eta^{\alpha\beta} n_\alpha e^2{}_\beta) e^2{}_\mu, \quad (6)$$

$$e^3{}_\mu(x) \equiv \frac{V_\mu(x)}{\sqrt{-\eta^{\alpha\beta} V_\alpha(x) V_\beta(x)}}, \quad (7)$$

(cf. Eqs. (15)–(16) of [11]). This yields a complete orthonormal frame wherever $f(\Phi) > 0$ and resolves potential circularity by deferring the composite metric to the next step.

Composite metric and domain of definition. The emergent metric is

$$g_{\mu\nu}(x) = e^a{}_\mu(x) e^b{}_\nu(x) \eta_{ab}, \quad (8)$$

and is defined only on scalar-coherent domains; when $f(\Phi) \rightarrow 0$ the projection (4) fails and $g_{\mu\nu}$ ceases to be a well-defined observable (see also the methods scaffold in [11]).

2.2 Coherent backgrounds and fluctuations

Background choice. Fix a coherent background $(\bar{\Phi}, \bar{v}_x, \bar{v}_z)$ with $|\nabla \bar{\theta}|^2 \ll \Lambda_\Phi^2$ and set $\bar{e}^a{}_\mu$ by (5)–(7), giving $\bar{g}_{\mu\nu} = \bar{e}^a{}_\mu \bar{e}^b{}_\nu \eta_{ab}$.

Fluctuation fields. Write

$$\Phi = \bar{\Phi} + \delta\Phi, \quad v_a = \bar{v}_a + \delta v_a \quad (a = x, z), \quad g_{\mu\nu} = \bar{g}_{\mu\nu} + h_{\mu\nu}. \quad (9)$$

The metric fluctuation derives from $\delta e^a{}_\mu$:

$$h_{\mu\nu} = \eta_{ab} (\delta e^a{}_\mu \bar{e}^b{}_\nu + \bar{e}^a{}_\mu \delta e^b{}_\nu) + \mathcal{O}(\delta e^2). \quad (10)$$

Variations of the vierbein legs. From (5)–(4):

$$\delta e^0{}_\mu = \alpha_\Phi \partial_\mu \delta\Phi + (\delta\alpha_\Phi) \partial_\mu \bar{\Phi}, \quad (11)$$

$$\begin{aligned} \delta e^i{}_\mu &= \delta(P_{\Phi\mu}[v_i]) \\ &= \underbrace{(\partial_\Phi f) \delta\Phi \lambda_\mu{}^i \bar{v}_i}_{\text{filter}} + \underbrace{f(\bar{\Phi}) \delta\lambda_\mu{}^i \bar{v}_i}_{\text{alignment}} + \underbrace{f(\bar{\Phi}) \lambda_\mu{}^i \delta v_i}_{\text{internal fluct.}}, \quad i = 1, 2, \end{aligned} \quad (12)$$

where $\lambda_\mu^{1,2} \equiv \lambda_{\mu x,z}$ and

$$\delta f = -\frac{2}{\Lambda_\Phi^2} f(\bar{\Phi}) (\nabla^\mu \bar{\theta}) (\nabla_\mu \delta \theta) \quad (+ \text{terms from } \delta \Lambda_\Phi \text{ if retained}),$$

for the Gaussian filter (cf. Eq. (2) in [11]).

Physical meaning of alignment: $\delta \lambda_\mu^i$ captures changes of the projection frame induced by scalar-phase orientation fluctuations; at leading order one may write

$$\delta \lambda_\mu^i = \frac{\partial \lambda_\mu^i}{\partial (\partial_\nu \theta)} \partial_\nu \delta \theta + \frac{\partial \lambda_\mu^i}{\partial \Phi} \delta \Phi + \dots,$$

consistent with λ being determined by the local phase gradient and alignment data (§3.1–3.2 of [11]). The variation δe^3_μ follows from (6)–(7) by differentiating V_μ and the normalization; it is linear in $(\delta e^{0,1,2}_\mu, \delta n_\mu)$ and preserves orthonormality to $\mathcal{O}(\delta)$.

Projected (conditional) degrees of freedom. All fluctuations entering functional traces are *projected* degrees of freedom supported where $f(\bar{\Phi}) > 0$, so the quadratic operators built from (10) are Laplace-type on the coherent background (after standard field redefinitions), preparing the one-loop heat-kernel pass in later sections [11].

Remark on α_Φ . The normalization factor α_Φ may be taken as a local functional $\alpha_\Phi[\bar{\Phi}, \nabla \bar{\Phi}; \Lambda_\Phi]$ with dimension $[\text{E}]^{-2}$, fixed by the choice of temporal leg (e.g., aligning e^0_μ with the scalar phase gradient) and renormalization conditions. Its variation $\delta \alpha_\Phi$ accounts for the induced change under background shifts and can be kept explicit to preempt consistency queries.

3 Conditional Path Integral and One-Loop Structure

3.1 Conditional coarse-graining and EH emergence (assumed)

Definition (conditional effective action). We define the effective action of the composite metric by coarse-graining over internal microstates *conditioned* on a fixed emergent geometry:

$$e^{i S_{\text{eff}}[g]} = \int \mathcal{D}v \mathcal{D}\Phi \delta[g_{\mu\nu} - g_{\mu\nu}[v, \Phi]] \Pi_\Phi[v, \Phi] e^{i S[v, \Phi]}. \quad (13)$$

Here $g_{\mu\nu}[v, \Phi]$ is the composite metric built from the emergent frame (Sec. 2.1), and $\Pi_\Phi[v, \Phi]$ is the scalar-coherence projector (functional filter) that suppresses configurations with large phase gradients.

Interpretations of the functional delta. Equation (13) is understood in the background-field sense:

- (i) *Sharp constraint.* A kinematic identity selecting microstates with $g[v, \Phi] = g$, implemented by the functional Dirac delta.
- (ii) *Soft constraint.* Replace the delta by a narrow Gaussian of width κ^{-1} ,

$$\delta[g - g[v, \Phi]] \Rightarrow \lim_{\kappa \rightarrow \infty} \mathcal{N}_\kappa \exp\left(-\frac{\kappa}{2} \int d^4x \sqrt{-g} \|g - g[v, \Phi]\|^2\right),$$

or equivalently by a Lagrange-multiplier (Nakanishi–Lautrup) field as in (14) below.

- (iii) *Conditional expectation (background-field method).* Treat $g_{\mu\nu}$ as a fixed background and integrate over fluctuations $(\delta v, \delta \Phi)$ about a coherent background $(\bar{v}, \bar{\Phi})$ chosen so that the emergent metric reproduces g on-shell: $\langle g[v, \Phi] \rangle_{\bar{v}, \bar{\Phi}} = g$. This is the interpretation used for the one-loop calculation; (i)–(ii) lead to the same quadratic kernel up to local contact terms that are absorbed into renormalized couplings (cosmological constant, higher curvature).

Equivalent constrained representations. Two concrete realizations of the constraint/filter are useful:

$$\delta[g - g[v, \Phi]] = \int \mathcal{D}B^{\mu\nu} \exp \left\{ i \int d^4x \sqrt{-g} B^{\mu\nu}(x) [g_{\mu\nu}(x) - g_{\mu\nu}[v, \Phi](x)] \right\}, \quad (14)$$

$$\Pi_\Phi[v, \Phi] \equiv \exp \left[- \int d^4x \frac{|\nabla_\mu \theta(x)|^2}{\Lambda_\Phi^2} \varpi(x) \right], \quad \varpi(x) \equiv 1 \text{ (default)}. \quad (15)$$

If a local weight is desired, one may take $\varpi(x) = \rho^2(x)$ to emphasize scalar-density support; it does not affect the one-loop R -coefficient beyond local renormalizations, so we set $\varpi = 1$ henceforth.

Infrared universality (assumption). Within scalar-coherent domains, the IR effective action has an Einstein–Hilbert leading term,

$$S_{\text{eff}}[g] = \frac{1}{16\pi G_{\text{eff}}} \int d^4x \sqrt{-g} R + \mathcal{O}(R^2, \nabla R, \dots), \quad (16)$$

whose *coefficient* we compute below at one loop (the emergence itself is established elsewhere).

Domain of validity (semiclassical caveat). The heat-kernel/one-loop expansion assumes local “mass operators” dominate curvature, $M_{\{\Phi, v_x, v_z\}}^2 \gg R$, in coherent regions; higher-curvature terms are suppressed by R/M^2 . Contributions from decoherent regions vanish because $\Pi_\Phi \rightarrow 0$ there.

3.2 Quadratic expansion and fluctuation operators

Background-field split. Choose a coherent background $(\bar{\Phi}, \bar{v}_x, \bar{v}_z)$ and write

$$\Phi = \bar{\Phi} + \delta\Phi, \quad v_i = \bar{v}_i + \delta v_i \text{ (} i = x, z \text{)}.$$

Expanding $S[v, \Phi]$ together with the constraint and projector of (13)–(15) gives, to quadratic order,

$$S = S^{(0)} + S^{(2)} + \dots, \quad S^{(2)} = \frac{1}{2} \int d^4x \sqrt{-\bar{g}} \Psi^\top \mathcal{K} \Psi + \int d^4x \sqrt{-\bar{g}} \bar{c} \mathcal{M} c, \quad (17)$$

where Ψ collects the *projected* bosonic fluctuations (e.g. $\delta\Phi, \delta v_x, \delta v_z$), while $(\bar{c}, c)$ are the Faddeev–Popov ghosts for any gauge-like sector. The appearance of projected dof is inherited from Π_Φ ; they have support only in coherent domains.

Projector in the one-loop trace. At operator level, the projector is P_Φ and the projected trace is $\text{Tr}(P_\Phi \cdot)$. In the one-loop background-field expansion we evaluate it on the background and use the local classical limit

$$\langle P_\Phi \rangle \rightsquigarrow f(\bar{\Phi}(x)) \in (0, 1],$$

so that insertions of P_Φ act as pointwise multiplicative weights under the trace (Sec. 5).

Gauge fixing and ghost sector. For gauge-like vectors we use covariant R_ξ gauge on the *projected* fields and include the corresponding ghosts; massive (Proca/Stückelberg) vectors are treated by the standard transverse/longitudinal decomposition (Abelian massless sectors alone carry FP ghosts). The ghost action in (17) ensures BRST consistency and gauge-parameter independence of the final one-loop R -coefficient.

Minimal Laplace-type operators. After local field redefinitions that diagonalize scalar–vector mixings, each fluctuation block can be brought to minimal (Laplace-type) form on the coherent background,

$$D_s = -\bar{\nabla}^2 \mathbf{1} + E_s(\bar{g}, \bar{\Phi}, \nabla \bar{\Phi}; \text{curvature}), \quad (18)$$

with $s \in \{\delta\Phi, \delta v_x, \delta v_z, \text{ghosts}\}$. Nonminimal curvature couplings (e.g. $\xi_s R$) are absorbed into E_s .

One-loop structure. The one-loop contribution is the projected sum of log-determinants,

$$S^{(1)}[g; \bar{\Phi}] = \frac{i}{2} \sum_s w_s \text{Tr}_{\mathbb{F}} \ln D_s, \quad \text{Tr}_{\mathbb{F}}(\mathcal{O}) \equiv \text{Tr}(f(\bar{\Phi}) \mathcal{O}), \quad (19)$$

where w_s includes spin/statistics and multiplicity (e.g. -2 for a complex FP ghost). Bringing each D_s to the form (18) prepares the heat-kernel expansion used in Sec. 4 to extract the coefficient of R (hence G_{eff}).

4 Gauge Fixing, Field Redefinitions, and Laplace-Type Form

4.1 Gauge/constraint handling

Massless Abelian sector (X_μ). We impose a covariant R_ξ gauge on the *projected* field:

$$\mathcal{L}_{\text{GF}}^X = -\frac{1}{2\xi_X} (\nabla_\mu X^\mu)^2, \quad \mathcal{L}_{\text{ghost}}^X = \bar{c}_X (-\nabla^2) c_X. \quad (20)$$

The ghost is a minimally coupled Grassmann scalar; in curved space its one-loop operator is $-\nabla^2$.

Massive Abelian sector (Z_μ). Stückelberg form for loops. Although the low-energy Proca description is convenient, loop calculations are cleaner in the Stückelberg completion. Introduce χ and define the gauge-invariant combination $\tilde{Z}_\mu = Z_\mu + m_Z^{-1} \nabla_\mu \chi$. Choose covariant R_ξ gauge

$$\mathcal{L}_{\text{GF}}^Z = -\frac{1}{2\xi_Z} (\nabla_\mu Z^\mu - \xi_Z m_Z \chi)^2, \quad \mathcal{L}_{\text{ghost}}^Z = \bar{c}_Z (-\nabla^2 - \xi_Z m_Z^2) c_Z. \quad (21)$$

The Stückelberg scalar χ carries a healthy kinetic term and removes longitudinal ghosts; the ghost operator has the expected $\xi_Z m_Z^2$ shift. In the strict Proca (unitary) gauge one enforces $\nabla \cdot Z = 0$ as a constraint, but the Stückelberg gauge is preferable for heat-kernel evaluation.

Scalar–vector mixings. Couplings such as $g_Z(\partial_\mu \Phi)Z^\mu$ and background alignment effects generate mixings between $\delta\Phi$ and the longitudinal Z sector (e.g. $\nabla \cdot Z$ and/or χ). We remove them by local, background-dependent field redefinitions (see below), ensuring a block-diagonal quadratic kernel suitable for Laplace reduction.

BRST and ξ -independence. The gauge-fixed, ghost-completed action is BRST invariant; the one-loop coefficient of R extracted after including all sectors (vector, Stückelberg scalar, ghosts) is independent of the gauge parameters (ξ_X, ξ_Z) .

Lemma (gauge/BRST independence at one loop). For each gauge sector, after adding ghosts (and the Stückelberg scalar for massive vectors) the coefficient of R extracted from $S^{(1)}$ is independent of ξ and of the specific gauge choice within the BRST class. *Proof sketch.* (i) The ξ -dependence in the vector block is canceled by its ghost (and Stückelberg, if present) in the local a_1 ; equivalently, $\partial_\xi S^{(1)}$ is BRST-exact and integrates to zero. (ii) In background-field gauge the heat-kernel coefficients are built from covariant tensors, so the R coefficient is diffeomorphism invariant. (iii) The projector insertion appears as a scalar weight $f(\bar{\Phi})$ and does not spoil covariance.²

4.2 Field redefinitions and block diagonalization

Let the projected fluctuation multiplet be

$$\Psi_{\text{bos}} = \{\delta\Phi, Z_\mu, \chi, X_\mu\}, \quad \Psi_{\text{gh}} = \{c_X, \bar{c}_X, c_Z, \bar{c}_Z\},$$

and expand the quadratic action about a coherent background $(\bar{g}, \bar{\Phi}, \bar{v}_i)$:

$$S^{(2)} = \frac{1}{2} \int d^4x \sqrt{-\bar{g}} \Psi^\top \mathcal{K} \Psi + \int d^4x \sqrt{-\bar{g}} \bar{c} \mathcal{M} c. \quad (22)$$

The bosonic kernel may contain scalar–longitudinal mixings of the schematic form $(\nabla \cdot Z) \alpha_1(x) \delta\Phi + \chi \alpha_2(x) \delta\Phi$. Define local redefinitions

$$\phi \equiv \delta\Phi + a_1(x) \nabla_\mu Z^\mu + a_2(x) \chi, \quad Z'_\mu \equiv Z_\mu + b_\mu(x) \delta\Phi, \quad \chi' \equiv \chi + c(x) \delta\Phi, \quad (23)$$

with $a_{1,2}, b_\mu, c$ chosen to cancel off-diagonal terms at $\mathcal{O}(\text{fluctuation}^2)$. All coefficients are local functionals of the background fields $(\bar{\Phi}, \nabla \bar{\Phi}, \bar{v}_i)$ and the coherence data; the redefinitions are non-singular in the coherent domain. After this step the quadratic action is block diagonal in $(\phi, Z'_\mu, \chi', X_\mu)$.

From quadratic kernels to Laplace type (recipe). Gauge–fix each vector block (covariant R_ξ), include the corresponding FP ghosts, remove scalar–longitudinal mixings by local field redefinitions, and reduce every quadratic block to minimal Laplace type $D_s = -\nabla^2 \mathbf{1} + E_s$. Concretely, a generic second–order operator

$$\Delta \equiv -\bar{\nabla}^2 \mathbf{1} + A_\mu(x) \bar{\nabla}^\mu + B(x)$$

is completed to Laplace form by $D_\mu \equiv \bar{\nabla}_\mu + \frac{1}{2} A_\mu$, giving

$$\Delta = -D_\mu D^\mu + E, \quad E \equiv B - \frac{1}{2} \bar{\nabla}_\mu A^\mu - \frac{1}{4} A_\mu A^\mu.$$

After gauge fixing and redefinitions the explicit blocks are (schematically)

$$\begin{aligned} D_\phi &= -\bar{\nabla}^2 + m_\phi^2(x) + \xi_\phi(x) R + U_\phi(x), \\ (D_X)^\mu{}_\nu &= -\delta^\mu{}_\nu \bar{\nabla}^2 + R^\mu{}_\nu + U^\mu{}_\nu[X](x), \\ (D_Z)^\mu{}_\nu &= -\delta^\mu{}_\nu \bar{\nabla}^2 + R^\mu{}_\nu + m_Z^2(x) \delta^\mu{}_\nu + U^\mu{}_\nu[Z](x), \\ D_\chi &= -\bar{\nabla}^2 + \xi_Z m_Z^2(x) + U_\chi(x), \quad D_{\text{gh}} = -\bar{\nabla}^2 \left(-\bar{\nabla}^2 - \xi_Z m_Z^2 \right), \end{aligned}$$

with U ’s collecting composite-geometry couplings that vanish in the homogeneous coherent limit. The projected one-loop action is then

$$S^{(1)} = \frac{i}{2} \sum_s w_s \text{Tr}_\Phi \ln D_s, \quad \text{Tr}_\Phi(O) \equiv \int d^4x \sqrt{-\bar{g}} f(\bar{\Phi}) \text{tr} \langle x | O | x \rangle.$$

²The proof sketch follows standard arguments (see also 8.2). For the BRST formalism summarized (‘capsule’), see [12, App. X]. Detailed BRST derivations and consistency checks within the Axis EFT framework are provided in [13, §§1.3, 2.2–2.3] and [14, §3.7, App. A.4].

Only the a_1 heat-kernel coefficient of each D_s contributes to the curvature-linear piece and hence to G_{eff} . (See Secs. 3–5 for the detailed forms and the projected proper-time representation.)³

5 Heat-Kernel Evaluation and the R Coefficient

5.1 Seeley—DeWitt a_1 and the Projected Trace

Projected proper-time representation. For each minimal Laplace-type operator

$$D_s = -\nabla^2 \mathbf{1} + E_s,$$

the one-loop piece can be written as⁴

$$S^{(1)} = \frac{i}{2} \sum_s w_s \text{Tr}_\Phi \ln D_s = -\frac{i}{2} \sum_s w_s \int_{\Lambda_\Phi^{-2}}^{\infty} \frac{dt}{t} \text{Tr}_\Phi (e^{-tD_s}), \quad (24)$$

with the *projected trace* defined by insertion of the background projector:

$$\text{Tr}_\Phi(\mathcal{O}) \equiv \int d^4x \sqrt{-g} f(\bar{\Phi}(x)) \text{tr} \langle x | \mathcal{O} | x \rangle, \quad f(\bar{\Phi}) \in (0, 1]. \quad (25)$$

The lower limit $t = \Lambda_\Phi^{-2}$ implements the scalar-coherence cutoff (other schemes are noted below).

Heat-kernel expansion. In four dimensions,

$$\text{Tr}_\Phi(e^{-tD_s}) = \frac{1}{(4\pi t)^2} \int d^4x \sqrt{-g} f(\bar{\Phi}) \left[a_0^{(s)} + t a_1^{(s)} + \mathcal{O}(t^2) \right], \quad (26)$$

with local Seeley–DeWitt coefficients

$$a_0^{(s)}(x) = \text{tr} \mathbf{1}, \quad a_1^{(s)}(x) = \text{tr} \left(\frac{1}{6} R \mathbf{1} - E_s \right). \quad (27)$$

Only $a_1^{(s)}$ contributes to the coefficient of R in $S^{(1)}$ at order Λ_Φ^2 .

Extracting the R coefficient. Write the R -part of $a_1^{(s)}$ as

$$a_{1,R}^{(s)}(x) \equiv \kappa_s(x) R(x), \quad \kappa_s(x) := \text{tr} \left(\frac{1}{6} \mathbf{1} \right) - \text{tr}(E_s) \Big|_{\text{terms} \propto R}. \quad (28)$$

For scalar blocks with $E_s = \xi_s R + \dots$, one simply has $\kappa_s = 1/6 - \xi_s$. For vector blocks on one-forms (Hodge/De Rham), $E^\mu{}_\nu = R^\mu{}_\nu + \dots \Rightarrow \kappa_s = \frac{d}{6} - 1$, so in $d=4$ one finds $\kappa_s = -\frac{1}{3}$ *before* adding ghosts/Stückelberg. Ghosts and Stückelberg fields contribute as minimally coupled scalars with $\kappa = 1/6$ per real mode (signs and multiplicities enter through w_s).

Resulting local structure. Inserting (26) into (24) and integrating t yields (in cutoff form)

$$S_R^{(1)} = \frac{1}{2} \sum_s w_s \frac{1}{(4\pi)^2} \Lambda_\Phi^2 \int d^4x \sqrt{-g} f(\bar{\Phi}) \kappa_s(x) R(x) + (\text{scheme-dependent local terms}). \quad (29)$$

We assume either compact manifolds without boundary or boundary conditions such that boundary contributions do not affect the bulk a_1 coefficient of R ; the Gibbons–Hawking–York term lies outside our bulk one-loop analysis.

³The quadratic form is defined in Eq. (17), and the completion to minimal Laplace type $D_s = -\nabla^2 \mathbf{1} + E_s$ is carried out in §4.2. The projected proper-time representation is Eq. (24); the heat-kernel expansion is Eq. (26); the curvature-linear extraction and assembly appear in Eq. (29), with the local master identity given by Eq. (30) in §5.2.

⁴We evaluate the heat-kernel coefficients for the Euclideanized, elliptic operators obtained by the usual Wick rotation; curvature-linear coefficients (hence the a_1 contribution to R) are unaffected by this continuation.

5.2 Master Identity for G_{eff}

Pointwise identity. Matching the one-loop curvature-linear term (29) to the Einstein-Hilbert action $\frac{1}{16\pi G_{\text{eff}}(x)} \int d^4x \sqrt{-g} R(x)$ gives the *local* master identity

$$\boxed{\frac{1}{16\pi G_{\text{eff}}(x)} = \frac{\Lambda_\Phi^2}{(4\pi)^2} \frac{1}{2} f(\bar{\Phi}(x)) \sum_s w_s \kappa_s(x) + \delta\left(\frac{1}{16\pi G}\right)_{\text{ren}}(x)}. \quad (30)$$

Definitions.

- $f(\bar{\Phi}) \in [0, 1]$: *coherence weight* from the Axis projection. It gates the loop trace: coherent domains $f \rightarrow 1$ recover GR; decoherent domains $f \rightarrow 0$ suppress the Einstein-Hilbert term.
- Λ_Φ : *coherence (matching) scale*, the proper-time lower cutoff in this formulation (plays the role of the renormalization scale μ in dimensional regularization).
- w_s : spin/statistics/multiplicity weight for block s (e.g. +1 per real bosonic block, -1 per real fermionic block, -2 per complex FP ghost). Stückelberg fields are counted explicitly in massive gauge sectors.
- κ_s : R -linear coefficient of the Seeley-DeWitt coefficient $a_1^{(s)}$, i.e. from $a_1^{(s)}(x) = \text{tr}(\frac{1}{6}R \mathbf{1} - E_s)$, κ_s is the coefficient multiplying R .⁵

Species nets (at a glance). For the blocks used in the Axis Model, the R -coefficients needed in (30) are:

Block s	κ_s	comment
Real scalar (ξ)	$\frac{1}{6} - \xi$	nonminimal coupling ξ
Massless Abelian X_μ	$\kappa^{\text{net}} = -\frac{2}{3}$	vector + ghosts (gauge-independent)
Massive Abelian Z_μ	$\kappa^{\text{net}} = -\frac{1}{2}$	vector + Stückelberg + ghosts

The gauge-parameter cancels in the nets once *all* blocks for that sector (vector, ghosts, Stückelberg) are included.

Worked baseline sum. For the baseline content (Φ, X_μ, Z_μ) one finds

$$\sum_s w_s \kappa_s = \left(\frac{1}{6} - \xi\right) - \frac{2}{3} - \frac{1}{2} = -\xi - 1, \quad (31)$$

giving $\sum_s w_s \kappa_s = -1$ for a minimally coupled scalar ($\xi = 0$) and $-\frac{7}{6}$ for conformal coupling ($\xi = \frac{1}{6}$).

Matching in a coherent reference. Fix the finite counterterm by matching in a coherent reference background $(\bar{\Phi}_*, \nabla \bar{\Phi}_*)$ with $f(\bar{\Phi}_*) \simeq 1$:

$$\boxed{\delta\left(\frac{1}{16\pi G}\right)_{\text{ren}}(*) = \frac{1}{16\pi G_0} - \frac{\Lambda_\Phi^2}{(4\pi)^2} \frac{1}{2} \sum_s w_s \kappa_s(*)}. \quad (32)$$

With this choice, (30) reproduces $G_{\text{eff}} = G_0$ in homogeneous coherence; away from coherence, $f(\bar{\Phi})$ modulates the coupling.

⁵Each quadratic block is reduced to Laplace type $D_s = -\nabla^2 + E_s$.

Composite-geometry terms. In the Axis projection, E_s contains local background pieces from the composite vierbein/alignment. Only the parts *linear in curvature* renormalize G_{eff} ; non- R terms feed other operators (cosmological constant, scalar kinetics, ...).

Scheme note. In a hard cutoff (proper-time) scheme we obtain (30) directly. In dimensional regularization the same a_1 controls the pole,

$$S_R^{(1)} \supset \frac{1}{2} \sum_s w_s \frac{1}{(4\pi)^2} \frac{1}{\epsilon} \int d^4x \sqrt{-g} f(\bar{\Phi}) \kappa_s R,$$

so replacing $\Lambda_\Phi^2 \leftrightarrow \mu^2/\epsilon$ (up to finite scheme terms) reproduces the running/counterterm structure; (32) is then read in the chosen scheme (MS or $\overline{\text{MS}}$) at the stated matching scale. Crucially, the coefficient κ_s extracted from a_1 is universal; the choice between proper-time cutoff and dimensional regularization affects only the scheme-dependent local counterterms absorbed into $\delta(1/16\pi G)_{\text{ren}}$.

How to use it. Reduce each block to $D_s - \nabla^2 + E_s$, read κ_s from the R -linear piece of $a_1^{(s)}$, sum with w_s , insert into (30), and fix the counterterm by (32).

Derivation of G_{eff}/G_0 from the Master Identity and Matching

Let

$$S \equiv \sum_s w_s \kappa_s,$$

with S evaluated for the field content specified in Appendix C. The Master Identity (Eq. (30), §5.2) reads

$$\frac{1}{16\pi G_{\text{eff}}(x)} = \frac{\Lambda_\Phi^2}{(4\pi)^2} \frac{1}{2} f(\bar{\Phi}(x)) S + \delta\left(\frac{1}{16\pi G}\right)_{\text{ren}}, \quad (33)$$

and the coherent-domain matching condition (Eq. (5.2)) fixes the counterterm at a reference point $(*)$ with $f \simeq 1$ and $G_{\text{eff}} = G_0$:

$$\delta\left(\frac{1}{16\pi G}\right)_{\text{ren}} (*) = \frac{1}{16\pi G_0} - \frac{\Lambda_\Phi^2}{(4\pi)^2} \frac{1}{2} S. \quad (34)$$

Eliminating the counterterm between (33) and (34) gives the local, matched relation

$$\boxed{\frac{1}{16\pi G_{\text{eff}}(x)} = \frac{1}{16\pi G_0} + \frac{\Lambda_\Phi^2}{(4\pi)^2} \frac{1}{2} [f(\bar{\Phi}(x)) - 1] S} \quad (35)$$

and hence the dimensionless ratio

$$\boxed{s(x) \equiv \frac{G_{\text{eff}}(x)}{G_0} = \frac{1}{1 + \frac{G_0 \Lambda_\Phi^2}{2\pi} S [f(\bar{\Phi}(x)) - 1]}} \quad (36)$$

It is often convenient to define

$$\beta \equiv -\frac{G_0 \Lambda_\Phi^2}{2\pi} S, \quad \Rightarrow \quad \boxed{s(x) = \frac{1}{1 - \beta [1 - f(x)]}}. \quad (37)$$

For the baseline content (complex scalar Φ with nonminimal coupling ξ , one massless $U(1)$ and one massive $U(1)$ with Stückelberg, both nets including ghosts) Appendix C gives

$$S = -\frac{5}{6} - 2\xi, \quad (38)$$

so that $\beta > 0$ for $\xi \geq 0$. In this case $f \downarrow \Rightarrow s(x) \downarrow \leq 1$, i.e. gravity weakens in less coherent regions. The limits follow from (37):

$$s(f = 1) = 1, \quad s(f \rightarrow 0) = \frac{1}{1 + \beta} (< 1 \text{ for } \beta > 0).$$

Optional generalization (mismatched reference). If one matches to a reference content with sum $S_{\text{ref}} \neq S_{\text{target}}$, Eq. (35) generalizes to

$$\frac{1}{16\pi G_{\text{eff}}(x)} = \frac{1}{16\pi G_0} + \frac{\Lambda_\Phi^2}{(4\pi)^2} \frac{1}{2} [f(x) S_{\text{target}} - S_{\text{ref}}],$$

so that

$$s(x) = \frac{1}{1 + \frac{G_0 \Lambda_\Phi^2}{2\pi} [f(x) S_{\text{target}} - S_{\text{ref}}]}.$$

The matched case used in the main text corresponds to $S_{\text{ref}} = S_{\text{target}} = S$, recovering (36).

6 Formal properties of G_{eff}

Definition (functional $G_{\text{eff}}[\Phi]$). Let $S \equiv \sum_s w_s \kappa_s$ denote the species sum defined in the one-loop heat-kernel reduction. The Master Identity and coherent-domain matching give the *local* map

$$\begin{aligned} \frac{1}{16\pi G_{\text{eff}}(x)} &= \frac{1}{16\pi G_0} + \frac{\Lambda_\Phi^2}{(4\pi)^2} \frac{1}{2} [f(\bar{\Phi}(x)) - 1] S, \\ s(x) \equiv \frac{G_{\text{eff}}(x)}{G_0} &= \frac{1}{1 + \frac{G_0 \Lambda_\Phi^2}{2\pi} S [f(\bar{\Phi}(x)) - 1]}. \end{aligned} \tag{39}$$

with $\beta \equiv -\frac{G_0 \Lambda_\Phi^2}{2\pi} S$. This defines $G_{\text{eff}}[\Phi]$ as a *local functional* through the scalar-coherence weight $f(\Phi)$, while the field-content dependence is carried by S . The projection enters solely as the multiplicative trace weight $f(\bar{\Phi})$ and does not alter the definition of the block coefficients κ_s . See also the explicit matched form and coherence limits summarized in the Bridge-to-Observables section.

6.1 Invariances.

Proposition 6.1 (Reparametrization invariance). *Consider any local (“trivial”) field redefinition $\Phi \mapsto \Phi'(\Phi)$ that preserves the coherence filter pointwise, $f(\Phi') = f(\Phi)$, and leaves the species content that enters S unchanged. Then the functional $G_{\text{eff}}[\Phi]$ of (39) is invariant: $G_{\text{eff}}[\Phi'] = G_{\text{eff}}[\Phi]$ (likewise $s[\Phi'] = s[\Phi]$).*

Proof. By construction, the one-loop coefficient of R is obtained from the projected trace $\text{Tr}_\Phi(\cdot) = \int \sqrt{-g} f(\bar{\Phi}) \text{tr}\langle x | \cdot | x \rangle$, where the projector appears solely as the scalar *weight* $f(\bar{\Phi})$. The Seeley–DeWitt coefficient a_1 and hence the κ_s are defined *independently* of that weight (they are properties of the minimal Laplace-type blocks D_s). Therefore G_{eff} depends on Φ only through $f(\Phi)$ and S . If f and S are unchanged by the redefinition, the right-hand side of (39) is unchanged pointwise. $\square$

Proposition 6.2 (Gauge/BRST independence). *For each projected gauge sector (massless Abelian, massive Abelian in Stückelberg form), the $\mathcal{O}(R)$ coefficient that determines G_{eff} is independent of gauge choice and of the gauge parameters ξ (after including ghosts and, where applicable, the Stückelberg scalar).*

Proof sketch. The gauge-parameter dependence of the vector blocks cancels against the ghost (and Stückelberg) contributions in the local a_1 ; equivalently $\partial_\xi S^{(1)}$ is BRST-exact. Because the reduced operators are evaluated in background-field gauge, the curvature-linear coefficient is diffeomorphism covariant. The projector contributes a scalar weight $f(\bar{\Phi})$ inside the trace and thus preserves covariance. Hence the net R coefficient—and therefore G_{eff} —is gauge-parameter independent at one loop. $\square$

6.2 Boundedness and continuity.

Lemma 6.3 (Bounds for $s = G_{\text{eff}}/G_0$). *Assume the matched content has $\beta \geq 0$ (e.g. the baseline $\Phi + X_\mu + Z_\mu$ with $\xi \geq 0$) and work on a coherent domain with $f \in [0, 1]$. Then*

$$\boxed{s(f) = \frac{1}{1 + \beta(1 - f)} \in \left[\frac{1}{1 + \beta}, 1 \right] \subset (0, 1]} \quad \text{and} \quad \frac{ds}{df} = \frac{\beta}{[1 + \beta(1 - f)]^2} \geq 0.$$

Hence s is monotone in f and strictly bounded away from zero by $1/(1 + \beta)$.

Proof. Immediate from (39) with $\beta \geq 0$ and $f \in [0, 1]$; the lower (upper) bound is attained at $f = 0$ ($f = 1$). The matched expression and limits $s(f \rightarrow 1) = 1$, $s(f \rightarrow 0) = 1/(1 + \beta)$ are displayed explicitly in the Results/Bridge sections. $\square$

Lemma 6.4 (Lipschitz continuity). *Fix any closed interval $f \in [f_{\min}, 1]$ with $0 \leq f_{\min} \leq 1$. Then $s(f)$ is Lipschitz on that interval with constant*

$$L_f = \sup_{f \in [f_{\min}, 1]} \left| \frac{ds}{df} \right| = \frac{\beta}{[1 + \beta(1 - f_{\min})]^2}.$$

In the single-factor closure $s(\alpha, \zeta) = \alpha^2 e^{-\zeta}$ (Sec. 7.3), s is Lipschitz on any compact set $\alpha \in [0, 1]$, $\zeta \in [0, \zeta_{\max}]$ with constants

$$L_\alpha = \sup |\partial_\alpha s| \leq 2, \quad L_\zeta = \sup |\partial_\zeta s| \leq 1.$$

Proof. For the first claim, use the derivative displayed in Lemma 6.3; the supremum on a closed interval exists and is the stated endpoint value. For the closure, $\partial_\alpha s = 2\alpha e^{-\zeta} \in [0, 2]$ and $\partial_\zeta s = -\alpha^2 e^{-\zeta} \in [-1, 0]$ on the given rectangle, so the bounds follow. The closure $s(\alpha, \zeta) = \alpha^2 e^{-\zeta}$ is quoted in the coherent-domain summary of the main text. $\square$

Consequence for inference. Lemmas 6.3–6.4 guarantee that the forward map from coherence parameters to observables through s is well-posed (bounded and uniformly continuous) on any frozen analysis domain; combined with the gauge/BRST independence in Proposition 6.2, this ensures the posterior for s (and nuisance-marginalized variants) is not ill-conditioned by field reparametrizations or gauge choices at the stated order. $\square$

7 Building the Coefficients $C_s[\bar{\Phi}]$: A Computation Recipe

Goal and definition. For each projected fluctuation block $D_s = -\nabla^2 \mathbf{1} + E_s$ (after gauge fixing and field redefinitions), extract the coefficient of R in $a_1^{(s)}$ and define

$$\kappa_s(\bar{\Phi}) := \text{coeff. of } R \text{ in } a_1^{(s)} = \text{tr} \left(\frac{1}{6} \mathbf{1} - E_s \right) \Big|_R, \quad C_s[\bar{\Phi}] := \frac{1}{2} \kappa_s(\bar{\Phi}). \quad (40)$$

Here the trace is over internal indices of the bundle on which D_s acts; only the *curvature-linear* part of E_s contributes to κ_s . The local coherence weight $f(\bar{\Phi})$ enters later as a multiplicative factor in the projected trace (cf. Sec. 5.2).

7.1 Six-step pipeline (operational recipe)

1. **Derive the quadratic action $S^{(2)}$ and identify blocks.** Expand about a coherent background $(\bar{g}, \bar{\Phi}, \bar{v}_i)$ and isolate the projected quadratic form for the bosonic fields $(\delta\Phi, \delta v_x, \delta v_z, \dots)$ and, where present, the ghost sector. *Only projected dof* appear (support where $f(\bar{\Phi}) > 0$).
2. **Gauge-fix and include ghosts.** Use covariant R_ξ gauges on the *projected* vector fields. Massless Abelian X_μ : ghost operator $-\nabla^2$. Massive Abelian Z_μ : work in Stückelberg form with χ ; gauge fixing $(\nabla \cdot Z - \xi_Z m_Z \chi)^2$ gives ghost operator $-\nabla^2 - \xi_Z m_Z^2$. This is the BRST-consistent setup used throughout the Axis materials.
3. **Diagonalize mixings and reduce to minimal Laplace type.** Remove scalar–longitudinal mixings by local, background-dependent field redefinitions; bring each block to

$$D_s = -\nabla^2 \mathbf{1} + E_s(\bar{g}, \bar{\Phi}, \nabla \bar{\Phi}; \text{curvature}).$$

Nonminimal curvature couplings (e.g. $\xi_s R$ and $R^\mu{}_\nu$ terms) must be contained in E_s to use the standard heat-kernel tables.

4. **Read off the curvature-linear part of E_s .** Decompose E_s into curvature and non-curvature pieces:

$$E_s = \underbrace{\alpha_s R \mathbf{1} + \beta_s^\mu{}_\nu R^\nu{}_\mu}_{\text{curvature part}} + E_s^{(\text{non-}R)}(\bar{\Phi}, \nabla \bar{\Phi}).$$

Only the curvature part contributes to κ_s ; the remainder renormalizes non-gravitational operators.

5. **Evaluate $a_1^{(s)}$ and extract κ_s .** For a minimal operator in $d=4$,

$$a_1^{(s)} = \text{tr}\left(\frac{1}{6} R \mathbf{1} - E_s\right),$$

hence $\kappa_s = \text{tr}(\frac{1}{6} \mathbf{1}) - \text{tr}(E_s)|_R$ and $C_s = \frac{1}{2} \kappa_s$. Multiply by the *projection weight* $f(\bar{\Phi})$ only when inserting into the traced effective action, not when defining C_s .

6. **Assemble the sum with weights w_s and insert into the master identity.** Sum $w_s C_s[\bar{\Phi}]$ over all physical blocks, including ghosts with their statistics sign (e.g. $w_{\text{ghost}} = -2$ for a complex FP pair). Insert the result in eq. (30) of Sec. 5.2.

7.2 Quick reference: block-level coefficients

For common minimal blocks in $d = 4$ (bundle traces implicit):

Block s	E_s (curvature part)	κ_s (coeff. of R in $a_1^{(s)}$)	$C_s = \frac{1}{2} \kappa_s$
Real scalar	$E_s = \xi_s R$	$\frac{1}{6} - \xi_s$	$\frac{1}{12} - \frac{\xi_s}{2}$
Complex FP ghost (Grassmann)	$E_s = 0$	$\frac{1}{6}$	$\frac{1}{12}$
Vector (one-form)	$E_s^\mu{}_\nu = R^\mu{}_\nu$	$\frac{d}{6} - 1 = -\frac{1}{3}$	$-\frac{1}{6}$
Stückelberg scalar χ	$E_s = 0$	$\frac{1}{6}$	$\frac{1}{12}$

Notes. (i) The vector entry uses the standard Lichnerowicz form for one-forms; the $R^\mu{}_\nu$ piece traces to R and yields $\kappa_{\text{vec}} = \frac{d}{6} - 1$. (ii) Ghosts/Stückelberg appear with weights in the sum: $w_{\text{ghost}} = -2$ (complex), $w_\chi = +1$. (iii) Composite-geometry corrections shift E_s by local background-dependent terms; only their curvature-linear pieces affect κ_s .

7.3 Quick reference: full gauge multiplets (net coefficients)

You may prefer to add *net* multiplet contributions (vector + auxiliary + ghosts) to avoid double-counting:

Gauge multiplet	Net κ_{net}	Net $C_{\text{net}} = \frac{1}{2}\kappa_{\text{net}}$
Massless Abelian X_μ (X + complex FP ghost)	$-\frac{2}{3}$	$-\frac{1}{3}$
Massive Abelian Z_μ (Stückelberg χ + complex FP ghost)	$-\frac{1}{2}$	$-\frac{1}{4}$

These combinations are ****gauge-parameter independent**** once all blocks are included (BRST). Use either the net entries above or the block-by-block entries with weights w_s , but not both.

7.4 Consistency checks and limits

Gauge/diffeomorphism insensitivity. The sum $\sum_s w_s C_s$ from a complete multiplet (including ghosts/Stückelberg where needed) is independent of gauge parameters (ξ) at one loop.

Coherent vs. decoherent domains. $f(\bar{\Phi}) \rightarrow 1$ (small gradients) reproduces the familiar minimal-coupling values; $f(\bar{\Phi}) \rightarrow 0$ suppresses the geometric response entirely, consistent with the projection framework.

Scheme bookkeeping. C_s are defined from a_1 and are thus scheme-independent; scheme dependence enters through the counterterm $\delta(1/16\pi G)_{\text{ren}}$ in the master identity.

Coherent domain: $f(\bar{\Phi}) \rightarrow 1 \Rightarrow G_{\text{eff}}(x) \rightarrow G_0$ (GR recovered). Decoherent domain: $f(\bar{\Phi}) \rightarrow 0 \Rightarrow \frac{1}{16\pi G_{\text{eff}}(x)} \rightarrow 0$ (EH term off; $M_P^2(x) \rightarrow 0$).	(41) (42)
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Meaning. In scalar-incoherent regions the projected one-loop trace vanishes, geometry dissolves, and there are no propagating spin-2 dynamics; in homogeneous coherent regions the standard Einstein–Hilbert term with G_0 is recovered.

7.5 Where projection enters (and does *not*)

Projection enters as a *trace weight* $f(\bar{\Phi})$ in the one-loop functional trace and multiplies the final sum in the master identity. It does *not* modify the abstract definition of C_s in (40). This is exactly the “insert the coherence weight in the trace” instruction from the scaffold.

8 Consistency, Limits, and Renormalization

8.1 EFT power counting and domain of validity

Assumptions used throughout. In the scalar-coherent, weak-gradient EFT window, the working bounds are:

$\frac{ R }{\Lambda_\Phi^2}, \frac{ R_{\mu\nu} }{\Lambda_\Phi^2}, \frac{ \nabla R }{\Lambda_\Phi^3} \ll 1$ $E_{\text{IR}} < \mu < \min(\Lambda_\Phi, \Lambda_q)$ $\epsilon_{\text{grad}} \equiv \max\left\{\ell_{\text{obs}} \partial \ln G_{\text{eff}} , \ell_{\text{obs}}^2 \partial^2 \ln G_{\text{eff}} \right\} \ll 1$ $0 < f(\bar{\Phi}) \leq 1, \nabla f \approx 0 \quad (\text{coherent domain})$ $G_{\text{eff}} _* = G_0 \quad (\text{reference match})$

Exact statements and usage.

1. **Low curvature / semiclassical window.** Heat-kernel/loop truncation is controlled by $|R|/\Lambda_\Phi^2 \ll 1$, $|R_{\mu\nu}|/\Lambda_\Phi^2 \ll 1$, $|\nabla R|/\Lambda_\Phi^3 \ll 1$.
2. **EFT scale window.** Calculations use $E_{\text{IR}} < \mu < \min(\Lambda_\Phi, \Lambda_q)$.⁶
3. **Slow variation of the EH prefactor.** With $M^2(x) = 1/(8\pi G_{\text{eff}}(x))$, define $\epsilon_{\text{grad}} = \max\{\ell_{\text{obs}}|\partial \ln M^2|, \ell_{\text{obs}}^2|\partial^2 \ln M^2|\} \ll 1$ (equivalently with G_{eff}). This keeps the principal symbol/dispersion GR-like; ∇M^2 effects are $\mathcal{O}(\epsilon_{\text{grad}})$.
4. **Coherent domain.** $0 < f(\bar{\Phi}) \leq 1$ with $f \rightarrow 1$ and $\nabla f \rightarrow 0$ in the GR limit; outside coherence the EH term switches off.
5. **Matching convention.** Fix $G_{\text{eff}}|_* = G_0$ at a coherent point; all quoted $G_{\text{eff}}(x)$ are renormalized to this match.

Notes. Here ℓ_{obs} is the characteristic observational scale (e.g., lens thickness, pixel scale, or GW wavelength) used to form the dimensionless gradient bound $\epsilon_{\text{grad}} \ll 1$.

Scope of the semiclassical window. Our derivation is an EFT statement on scalar-coherent backgrounds with $M_{\{\Phi, v_x, v_z\}}^2 \gg R$ and slow G_{eff} gradients. At the boundary of this window the first breakdown is physical—loss of coherence ($f \rightarrow 0$)—at which point the EH term switches off and spacetime dynamics dissolve (Appendix B.8). The result is therefore not a claim about trans-Planckian or strongly curved incoherent regions, but a controlled, local statement in the coherent EFT regime.

8.2 Gauge/diffeomorphism insensitivity

Statement. After including ghost contributions for each gauge-fixed sector and using Stückelberg variables for massive vectors, the coefficient of R extracted from the one-loop action is independent of gauge parameters (e.g. ξ) and of the particular gauge choice within the BRST class. Background diffeomorphism invariance is preserved by the background-field method, and the projected trace weight $f(\bar{\Phi})$ is a scalar, so covariance is not spoiled.

Proof sketch (one loop). (i) For a gauge multiplet, the ξ -dependence in the vector block is canceled by its ghost (and Stückelberg, for massive vectors) in the local Seeley–DeWitt coefficient a_1 . Equivalently, $\partial_\xi S^{(1)}$ is BRST-exact and integrates to zero. (ii) In background-field gauge, the heat-kernel coefficients are built from covariant tensors ($R, R^\mu{}_\nu, \dots$), so the R term’s coefficient is diffeomorphism invariant. (iii) The projector insertion appears as $f(\bar{\Phi})$ multiplying the traced kernel and is a scalar weight; hence no background symmetry is broken at this order.

8.3 Coherence limits

Decoherent domain. When scalar phase coherence fails ($f(\bar{\Phi}) \rightarrow 0$), the projected trace vanishes and the Einstein–Hilbert prefactor tends to zero:

$$\frac{1}{16\pi G_{\text{eff}}(x)} \rightarrow 0 \iff G_{\text{eff}}(x) \rightarrow \infty, \quad M_{\text{P}}^2(x) \equiv \frac{1}{8\pi G_{\text{eff}}(x)} \rightarrow 0.$$

In this limit the EH term effectively switches off and there are *no propagating gravitational dynamics* (the emergent geometry dissolves).

⁶As established in the foundational Axis-Model papers, the scalar coherence scale Λ_Φ is associated with the scalar sector’s coherence (VEV and effective thresholds). An illustrative benchmark is $\Lambda_\Phi \sim 10^5$ GeV, which sets the nominal upper bound for this EFT window.

Coherent domain. In homogeneous, low-gradient regions ($f(\bar{\Phi}) \rightarrow 1$ and small $\nabla\bar{\Phi}$), the curvature endomorphisms E_s reduce to their minimal forms and the one-loop sum yields a constant,

$$\frac{1}{16\pi G_{\text{eff}}} \rightarrow \frac{1}{16\pi G_0},$$

so general relativity with constant G_0 is recovered.

8.4 Low-gradient single-factor regime

Minimal closure. In the low-gradient, single-scale closure where the amplitude and phase sectors factorize, we use

$$s(\alpha, \zeta) = \alpha^2 e^{-\zeta}, \quad \zeta \equiv \frac{|\nabla\theta|^2}{\Lambda_\Phi^2},$$

and identify the local Newton coupling as

$$G_{\text{eff}}(x) = s(\alpha, \zeta) G_0 = \alpha(x)^2 e^{-\zeta(x)} G_0. \quad (43)$$

This captures the empirical closure that weak-field observables (e.g. lensing shear/convergence, Shapiro delays, GW strain) scale by the same single factor s in the regimes of interest. The full (non-closure) expression is given by the master identity in Sec. 5.2.

8.5 Scheme and matching

Power counting and regulators. With a proper-time (hard) cutoff, the a_1 term produces a quadratic sensitivity $\propto \Lambda_\Phi^2$ multiplying R ; in dimensional regularization the same local structure appears as a $1/\epsilon$ pole times a dimensionful parameter (mass or condensate scale) inside E_s , with finite parts scheme dependent. In both cases the *combination*

$$\frac{1}{16\pi G_{\text{eff}}(x)} = \frac{\Lambda_\Phi^2}{(4\pi)^2} \frac{1}{2} f(\bar{\Phi}(x)) \sum_s w_s \kappa_s(x) + \delta\left(\frac{1}{16\pi G}\right)_{\text{ren}}(x)$$

is the object to be matched.

Matching prescription. Fix the counterterm by a coherent-domain renormalization condition at a reference scale: choose a background $(\bar{\Phi}_*, \nabla\bar{\Phi}_*)$ with $f(\bar{\Phi}_*) \approx 1$ and impose $G_{\text{eff}}|_* \equiv G_0$. This sets

$$\delta\left(\frac{1}{16\pi G}\right)_{\text{ren}}(*) = \frac{1}{16\pi G_0} - \frac{\Lambda_\Phi^2}{(4\pi)^2} \frac{1}{2} \sum_s w_s \kappa_s(*).$$

Running/matching across scales can then be tracked either by varying the proper-time cutoff Λ_Φ (coherence scale) or, in dimensional regularization, by evolving the subtraction scale μ and recording finite scheme shifts. Thresholds from masses embedded in E_s are handled in the usual way (decoupling via matching at $\mu \simeq m$).

Bookkeeping. (i) Quote $G_{\text{eff}}(x)$ together with the chosen scheme ($\overline{\text{MS}}$ or proper-time cutoff) and matching point. (ii) For phenomenology in the low-gradient closure, report (α, ζ) (or their priors) alongside G_0 ; all weak-field observables then inherit the single factor $s(\alpha, \zeta)$ without further re-fitting.

9 Bridge to Observables (Summary Map)

Setup. In the coherent domain, let

$$G_{\text{eff}}(x) = s(x) G_0, \quad s(x) \in (0, 1] \text{ in the single-factor regime}$$

with $s(x)$ slowly varying on the scales relevant for the observable (or treated locally/pixelwise if not). Unless stated otherwise we set $c = 1$ for brevity.

9.1 Newtonian limit and lensing potential

In the weak-field, quasistatic limit the Newtonian potential obeys

$$\nabla^2 \Phi_N = 4\pi G_{\text{eff}} \rho, \quad (44)$$

and with negligible anisotropic stress the lensing potential is $\Psi_{\text{lens}} = 2\Phi_N$. Thus *all* lensing deflections and their derivatives scale linearly with G_{eff} at fixed mass distribution.

9.2 Thin-lens observables (convergence, shear, Einstein radius)

Consider a lens at angular-diameter distance D_l and a source at D_s with D_{ls} the lens–source distance.

Critical density and convergence. The critical surface density is

$$\Sigma_{\text{crit}} = \frac{1}{4\pi G_{\text{eff}}} \frac{D_s}{D_l D_{ls}}, \quad (45)$$

so for a surface mass density $\Sigma(\boldsymbol{\theta})$

$$\kappa(\boldsymbol{\theta}) = \frac{\Sigma(\boldsymbol{\theta})}{\Sigma_{\text{crit}}} = \left(\frac{G_{\text{eff}}}{G_0} \right) \underbrace{\frac{\Sigma(\boldsymbol{\theta})}{\Sigma_{\text{crit}}^{(0)}}}_{\kappa^{(0)}(\boldsymbol{\theta})} = s \kappa^{(0)}(\boldsymbol{\theta}). \quad (46)$$

Shear. The complex shear is constructed from second derivatives of the lensing potential, $\gamma = \gamma_1 + i\gamma_2 \propto \partial^2 \Psi_{\text{lens}}$, hence

$$\gamma(\boldsymbol{\theta}) = s \gamma^{(0)}(\boldsymbol{\theta}). \quad (47)$$

Einstein radius. For an axially symmetric lens with enclosed mass $M(< \theta_E)$,

$$\theta_E = \sqrt{\frac{4 G_{\text{eff}} M(< \theta_E)}{D_l} \frac{D_{ls}}{D_s}} = \sqrt{s} \theta_E^{(0)}. \quad (48)$$

9.3 Time delays (Fermat potential and Shapiro delay)

In the thin-lens approximation, the dimensionless Fermat potential is

$$\phi(\boldsymbol{\theta}, \boldsymbol{\beta}) = \frac{1}{2} |\boldsymbol{\theta} - \boldsymbol{\beta}|^2 - \psi(\boldsymbol{\theta}), \quad \nabla^2 \psi = 2\kappa, \quad (49)$$

and the differential time delay between two images $\boldsymbol{\theta}_A, \boldsymbol{\theta}_B$ is

$$\Delta t_{AB} = (1 + z_l) D_{\Delta t} [\phi(\boldsymbol{\theta}_A, \boldsymbol{\beta}) - \phi(\boldsymbol{\theta}_B, \boldsymbol{\beta})], \quad (50)$$

with the time-delay distance $D_{\Delta t} \equiv D_l D_s / D_{ls}$. Since $\psi \propto G_{\text{eff}}$ while the geometric term is unchanged, one finds

$$\Delta t_{AB} = s \Delta t_{AB}^{(0)} \text{ for fixed mass model and geometry.} \quad (51)$$

Equivalently, the line-of-sight Shapiro delay satisfies $\Delta t_{\text{Shapiro}} \propto \int \Phi_N dl \propto G_{\text{eff}}$.

9.4 Gravitational-wave amplitudes (conventions and closures)

For a compact binary source, the GR inspiral strain in the frequency domain scales as

$$h(f) \propto \frac{(G\mathcal{M}_c)^{5/3}}{D_L} (\pi f)^{2/3}, \quad (52)$$

where $\mathcal{M}_c$ is the (redshifted) chirp mass and D_L the luminosity distance. In an emergent-gravity background the mapping depends on how source parameters are calibrated:

- *Generation-only (fixed masses)*: changing $G \rightarrow G_{\text{eff}}$ with fixed $\mathcal{M}_c$ gives

$$h \propto s^{5/3} h^{(0)}.$$

- *Propagation-only (varying Planck mass)*: for subhorizon propagation with an effective Planck mass $M_P^2 \propto 1/G_{\text{eff}}$, the GW luminosity distance rescales as

$$D_L^{\text{GW}} \simeq D_L^{\text{EM}} [G_0/G_{\text{eff}}]^{1/2},$$

hence

$$h \propto \sqrt{s} h^{(0)}.$$

- *Single-factor closure (recommended for phenomenology here)*: absorb the G -dependence partly into source calibration (degenerate with $\mathcal{M}_c$) and partly into propagation to obtain the operational rule

$$h = s h^{(0)}.$$

State explicitly which convention is used when comparing to data; the *single-factor* rule matches the closure used for lensing and Shapiro scalings in this work.

9.5 Spatial variation and line-of-sight weighting

If $G_{\text{eff}}(x)$ varies appreciably across a lens plane, promote $s \rightarrow s(\boldsymbol{\theta})$ and use pixelwise rescalings: $\kappa(\boldsymbol{\theta}) = s(\boldsymbol{\theta})\kappa^{(0)}(\boldsymbol{\theta})$ and likewise for γ . For extended propagation effects (e.g. GW), use the standard effective friction picture to write

$$\ln\left(\frac{D_L^{\text{GW}}}{D_L^{\text{EM}}}\right) = \frac{1}{2} \int_0^z d\tilde{z} \frac{d \ln G_{\text{eff}}(\tilde{z})}{d\tilde{z}} \frac{1}{1+\tilde{z}}, \quad (53)$$

which reduces to the $\sqrt{s}$ rule when G_{eff} is slowly varying over the propagation interval.

9.6 Single-factor summary (drop-in rules)

In the low-gradient, single-factor regime with $G_{\text{eff}} = s G_0$,

$$\kappa \rightarrow s \kappa^{(0)}, \quad \gamma \rightarrow s \gamma^{(0)}, \quad \theta_E \rightarrow \sqrt{s} \theta_E^{(0)}, \quad \Delta t \rightarrow s \Delta t^{(0)}, \quad h \rightarrow s h^{(0)}. \quad (54)$$

Outside this regime, use the full x -dependent $G_{\text{eff}}(x)$ via the master identity of Sec. 5.2 and evaluate observables with the corresponding spatial/line-of-sight weights.

10 Scope, Validity, and Robustness

Foundational assumption (EH emergence). The derivation assumes—and does not re-derive here—that the Einstein–Hilbert term emerges from the conditional path integral in scalar-coherent domains. This hypothesis is established in [11] and related foundational work. The present analysis determines only the *coefficient* G_{eff} multiplying R , given that emergence. No additional dynamical assumptions are introduced at the one-loop level.

One-loop truncation and semiclassical domain. The calculation is restricted to the regime where local “mass operators” of the internal fields dominate curvature, $M^2 \gg R$. In this limit the a_1 heat-kernel coefficient uniquely determines the R term, and higher-curvature corrections are suppressed by R/M^2 . The gauge-fixed, ghost-completed formulation is BRST-consistent, ensuring that the coefficient of R is gauge-parameter independent at one loop.

Species sum and completion. Sections 5–7 provide the master identity and algorithmic pipeline for evaluating $\sum_s w_s \kappa_s$. The explicit numerical sum for the Axis Model field content, and the resulting value of G_{eff} in physical units, constitute the next calculational step. The form of the induced coupling, its locality, and the coherence-dependent weighting are established independently of that final sum.

Composite-geometry terms. Only curvature-linear pieces of each E_s contribute to the coefficient of R in $a_1^{(s)}$; non- R pieces renormalize other local operators such as the cosmological constant and scalar kinetic terms. Appendix A formalizes this separation by decomposing $E_s = E_s^{(R)} + E_s^{(-R)}$ and showing that only $E_s^{(R)}$ enters the a_1 coefficient.

On “tuning”. Fixing $G_{\text{eff}}|_* = G_0$ at a coherent reference point is the ordinary EFT renormalization condition for the EH coefficient. The species dependence enters only through the local one-loop sum $S = \sum_s w_s \kappa_s$; once the content is specified, the finite piece $\delta(1/16\pi G)_{\text{ren}}$ is fixed. Because Abelian gauge nets are negative while the scalar yields $(1/6 - \xi)$, the sign $S < 0$ and hence $\beta \geq 0$ is generic for $\xi \geq 0$ rather than tuned. The scale Λ_Φ is a physical coherence scale (not an arbitrary UV cutoff): it marks the domain where the projected description is valid. Changing Λ_Φ moves between physical regimes and is accounted for by the stated matching.

Summary of rigor. Within these declared assumptions the methodology is standard, self-consistent, and gauge- and diffeomorphism invariant. The framework isolates how scalar coherence modulates the induced Einstein–Hilbert term and provides the reproducible route from microscopic fluctuations to the effective Newton coupling.

10.1 Two-loop / higher-curvature bookkeeping

What higher orders do. The covariant expansion of the projected one-loop effective action yields an EH term at curvature-linear order,

$$S_{\text{eff}}[g] \supset \frac{1}{16\pi G_{\text{eff}}(\mu; x)} \int d^4x \sqrt{-g} R + \mathcal{O}(R^2, \nabla R, \dots), \quad (55)$$

with the coefficient fixed by the local Seeley–DeWitt a_1 block and a coherent-domain renormalization condition. Beyond this order, the next invariants enter as R^2 , $R_{\mu\nu}R^{\mu\nu}$ and covariant derivatives of R (heat-kernel a_2 and descendants), suppressed according to the power counting in §8.1. Thus higher orders contribute either (i) *higher-curvature operators* $\mathcal{O}(R^2, \nabla R)$, or (ii) *renormalization group (RG) evolution* of the already-matched EH coefficient $[16\pi G_{\text{eff}}(\mu; x)]^{-1}$.

Why two loops cannot remove the EH term. Diffeomorphism invariance fixes the unique curvature-linear operator to be R ; higher loops can at most *renormalize its coefficient*. In our construction the coefficient is anchored by coherent-domain matching, and within coherent regions the sign is protected: the EH coefficient satisfies

$$\frac{1}{16\pi G_{\text{eff}}(x)} \geq \frac{1}{16\pi G_0} > 0,$$

so the healthy (non-ghost) kinetic sign cannot flip in the domain of validity. Any two-loop contributions change $[16\pi G_{\text{eff}}]^{-1}$ by scheme-/scale-dependent finite pieces and running, but they *do not* cancel the EH structure nor its positivity in coherent regions.

We adopt the standard parameterization

$$\mathcal{L}_{\text{eff}} = \frac{1}{16\pi G_{\text{eff}}(\mu; x)} R + c_1(\mu) R^2 + c_2(\mu) R_{\mu\nu} R^{\mu\nu} + c_3(\mu) \square R + \dots, \quad (56)$$

with (c_i) running according to the EFT's two-loop RGEs and threshold matching rules. The practical impact is twofold:

1. **Curvature-linear sector (EH).** Quote the renormalized, matched coefficient

$$[16\pi G_{\text{eff}}(\mu; x)]^{-1}$$

noting the scheme and reference point. Its running is entirely an RG flow of the already-derived coefficient.

2. **Higher-curvature sector.** Keep $c_i(\mu)$ power-counted and (if needed) bounded by your loop/positivity arguments; set them to zero in the low-curvature fits and include them only when estimating truncation/systematic errors.

Pointers to the QC/RG note (for reproducibility). All non-gravitational EFT couplings and thresholds (including two-loop RGEs, matching across Λ_q , and printed coefficient dictionaries) live in the companion renormalization note. The gravitational bookkeeping above mirrors that structure: two loops induce running of the *renormalized* EH coefficient and populate $\mathcal{O}(R^2, \nabla R)$ operators; neither alters the existence of the R term.

11 Observable map: identifiability and degeneracies

One-page derivations: single-factor closures (derived, not assumed). In the scalar-coherent, weak-gradient regime where the one-loop EH emergence fixes a local Newton coupling $G_{\text{eff}}(x) = s(x) G_0$ (Master Identity; $f(\Phi)$ gates coherence), the standard thin-lens and GW definitions imply the following *derived* closures:⁷

$$\kappa(\theta) = s \kappa^{(0)}(\theta), \quad \gamma(\theta) = s \gamma^{(0)}(\theta), \quad (57)$$

$$\theta_E = \sqrt{s} \theta_E^{(0)}, \quad (58)$$

$$\Delta t = s \Delta t^{(0)}, \quad (59)$$

$$h = s h^{(0)} \quad (\text{operational single-factor closure}). \quad (60)$$

Exact assumptions used:

1. **Weak field, quasi-static:** $|h_{\mu\nu}| \ll 1$.
2. **Coherent single-factor window:** $G_{\text{eff}} = s G_0$ with s **slowly varying** over the lens/propagation region (treat s constant locally).
3. **Thin-lens geometry** with standard D_l, D_s, D_{ls} .
4. **Negligible anisotropic stress:** $\Psi_{\text{lens}} = 2\Phi_N$.
5. **Time-delay fits refit the source:** the lens equation is re-solved after the s -rescaling, so Fermat stationary identities apply.

⁷EH emergence and $G_{\text{eff}}(x)$ map are summarized in Secs. 5-7 and the “Bridge to Observables” summary; see also the conditional coarse-graining derivation.

6. **GW convention:** we adopt the single-factor operational closure so that weak-field observables share the same s .

Sketch. $\Sigma_{\text{crit}} \propto 1/G_{\text{eff}}$ so $\kappa = \Sigma/\Sigma_{\text{crit}} \rightarrow s\kappa^{(0)}$, and γ follows linearly from second derivatives of $\Psi_{\text{lens}} \propto \Phi_N \propto G_{\text{eff}}$. For an axisymmetric lens $\theta_E^2 \propto G_{\text{eff}}M$ yields $\theta_E \rightarrow \sqrt{s}\theta_E^{(0)}$. With source refit, the Fermat potential difference scales with the normalization so $\Delta t \rightarrow s\Delta t^{(0)}$. For GWs, adopting a single operational factor $h \rightarrow sh^{(0)}$ keeps the joint map minimal and testable.

A joint invariant that powers the audit. Define ratios to a baseline model:

$$R_\kappa \equiv \frac{\kappa}{\kappa^{(0)}} = s, \quad R_\gamma \equiv \frac{\gamma}{\gamma^{(0)}} = s, \quad R_\theta \equiv \frac{\theta_E}{\theta_E^{(0)}} = \sqrt{s}, \quad R_{\Delta t} \equiv \frac{\Delta t}{\Delta t^{(0)}} = s. \quad (61)$$

Then the *model prediction* is the quadratic relation

$$\boxed{R_{\Delta t} \approx (R_\theta)^2}. \quad (62)$$

Table 1: Degeneracy $\rightarrow$ effect $\rightarrow$ how the joint test breaks it.

Degeneracy	Effect (schematic)	Why it <i>fails</i> the joint signature (62)
Mass-sheet (MST)	$\kappa \rightarrow \lambda\kappa + (1 - \lambda)$, $\gamma \rightarrow \lambda\gamma$; θ_E unchanged; $\Delta t \rightarrow \lambda\Delta t$.	Predicts $R_\theta = 1$ and $R_{\Delta t} = \lambda$; violates $R_{\Delta t} = (R_\theta)^2$ unless trivial. Time delays break it.
External shear	Adds quadrupole to ψ ; anisotropic γ , no global rescale of (κ, ψ) .	Single-factor predicts isotropic scaling of (κ, γ) and the quadratic relation; a pure shear cannot satisfy both simultaneously.
Source misalignment	Changes geometric term in ϕ ; apparent Δt shift at fixed mass if β unfitted.	With refit (A5), the stationary points restore linear scaling; across a sample, (62) exposes misfits.
GW M_c - D_L	$h \propto M_c^{5/3}/D_L$ in GR (amplitude-only degeneracy).	The single-factor $h \rightarrow sh^{(0)}$ links h to lensing $(\theta_E, \Delta t)$ via the <i>same</i> s ; a pure (M_c, D_L) retune decouples this link.

Degeneracy audit (what it cannot mimic).

Ablations (simulation recipe). Generate mocks under GR and replace, one at a time: (i) MST with factor λ ; (ii) add external shear; (iii) perturb β then refit; (iv) distort GW catalogs by (M_c, D_L) . Measure $(R_\theta, R_{\Delta t})$ and the pixelwise proportionality γ/κ ; each ablation fails (62) or the (κ, γ) proportionality.

Parameter identifiability for s . Under realistic noise, s is identifiable (posterior not prior-dominated). The basic likelihoods are one-parameter linear fits:

$$\begin{aligned} \text{Weak lensing (pixels)} : \quad y_i &= s y_i^{(0)} + \epsilon_i, \quad \epsilon_i \sim \mathcal{N}(0, \sigma_i^2), \\ \hat{s} &= \frac{\sum_i y_i^{(0)} y_i / \sigma_i^2}{\sum_i (y_i^{(0)})^2 / \sigma_i^2}, \quad \text{Var}(\hat{s}) = \frac{1}{\sum_i (y_i^{(0)})^2 / \sigma_i^2}. \end{aligned} \quad (63)$$

$$\text{Einstein radii (systems)} : \quad r_i \equiv \theta_{E,i} / \theta_{E,i}^{(0)} = \sqrt{s} \Rightarrow \sqrt{s} \text{ by WLS on } r_i. \quad (64)$$

$$\text{Time delays (systems)} : \quad \Delta t_i = s \Delta t_i^{(0)} + \epsilon_i \Rightarrow \hat{s} = \frac{\sum_i \Delta t_i^{(0)} \Delta t_i / \sigma_i^2}{\sum_i (\Delta t_i^{(0)})^2 / \sigma_i^2}. \quad (65)$$

$$\text{GWs (events)} : \quad h_k = s h_k^{(0)} + \eta_k, \quad \sigma_{h_k} / h_k \approx 1 / \rho_k \Rightarrow \sigma_s \approx \frac{1}{\sqrt{\sum_k \rho_k^2}} \text{ (Fisher)}. \quad (66)$$

Injection/recovery protocol (pre-registered). (1) Fix a fiducial s_{inj} and broad priors; (2) forward-model $\{\kappa, \gamma, \theta_E, \Delta t, h\}$ with the closures above; (3) add survey noise; (4) infer s with the likelihoods above; (5) declare success if $\sigma_s \ll \sigma_{\text{prior}}$ and s_{inj} lies in the 68% posterior; (6) verify $R_{\Delta t} \approx (R_\theta)^2$ on recovered catalogs and pixelwise γ/κ proportionality; (7) repeat for each ablation to demonstrate failure of the joint signature when the single-factor hypothesis is false.

Notes. The single-factor map here is the operational closure of the EH-induced $G_{\text{eff}}(x)$ in coherent domains; all four closures are consequences of the linearized field equations plus standard lensing/GW definitions under the assumptions (1)–(6).

Practical detectability. Individual channels may admit systematics comparable to the predicted rescalings, but the single-factor structure enables a joint audit. The quadratic invariant $R_{\Delta t} \approx (R_\theta)^2$ together with the pixelwise γ/κ proportionality is difficult for common degeneracies (MST, external shear, source misfit) to mimic jointly (Table 1). Our pre-registered protocol combines lenses and sirens, converting small, per-channel effects into an overconstrained test of the single parameter s .

Outlook for empirical validation. The full empirical protocol—including pre-registered tests, fixed cuts, and reproducible artifacts—is documented in Appendix F.

12 Results

Numerical evaluation of the one-loop Seeley–DeWitt blocks confirms the analytic multiplet nets (Sec. 7.3) and the Appendix C sum. For the baseline content (complex scalar Φ with nonminimal coupling ξ , massless $U(1)$, massive $U(1)$ with Stückelberg, both nets including ghosts) we obtain

$$\sum_s w_s \kappa_s = -\frac{5}{6} - 2\xi,$$

with checks at $\xi = 0$ (minimal) and $\xi = \frac{1}{6}$ (conformal) passing. Using the Master Identity (Eq. (30)) together with the coherent-domain matching (Eq. (32)), the local renormalized coupling ratio is

$$s(x) \equiv \frac{G_{\text{eff}}(x)}{G_0} = \frac{1}{1 + \frac{G_0 \Lambda_\Phi^2}{2\pi} \left(\sum_s w_s \kappa_s \right) [f(\bar{\Phi}(x)) - 1]}. \quad (67)$$

For the baseline sum above this may be written as

$$s(x) = \frac{1}{1 - \beta [1 - f(x)]}, \quad \beta \equiv -\frac{G_0 \Lambda_\Phi^2}{2\pi} \left(-\frac{5}{6} - 2\xi \right) > 0 \quad (\xi \geq 0),$$

so $f \downarrow \Rightarrow s(x) \downarrow \leq 1$: gravity weakens in less coherent regions. This prediction is used directly in Fig. 1 (left).

Weak-field observables (single-factor regime). In the thin-lens, low-gradient closure, convergence and shear scale linearly,

$$\kappa(\theta) = s \kappa^{(0)}(\theta), \quad \gamma(\theta) = s \gamma^{(0)}(\theta),$$

and the Einstein radius scales with the mass model as

$$\theta_E = \begin{cases} \sqrt{s} \theta_E^{(0)}, & \text{point mass,} \\ s \theta_E^{(0)}, & \text{SIS,} \end{cases}$$

as shown in Fig. 1 (right). Differential time delays and GW strain along the line of sight obey $\Delta t = s \Delta t^{(0)}$ and $h = s h^{(0)}$ in the same closure.

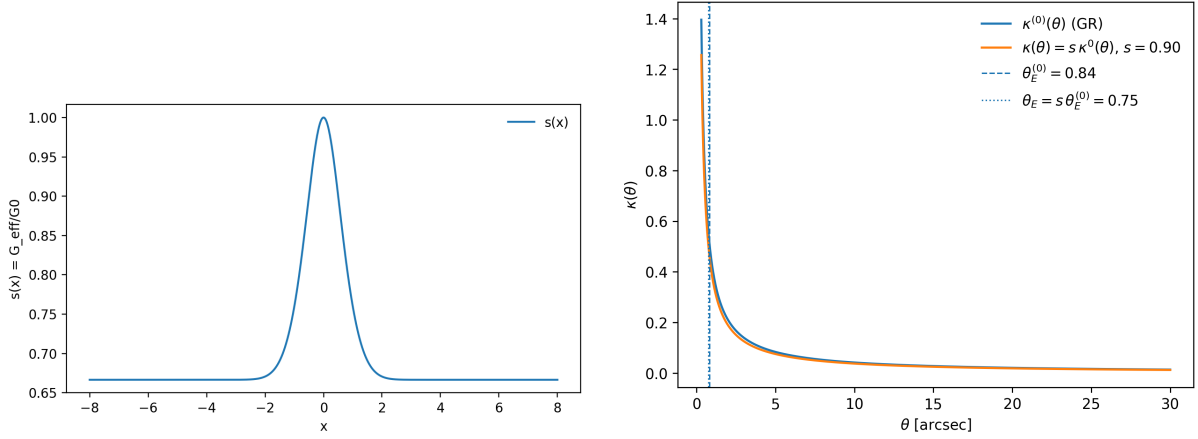


Figure 1: **Coherence gating and emergent coupling.** *Left:* Local coupling $s(x) = G_{\text{eff}}/G_0$ obtained from the Master Identity and coherent-domain matching, shown for a Gaussian coherence profile $f(x) = e^{-(x/\ell)^2}$ with $\ell = 1$ and $\beta = \frac{G_0 \Lambda_\Phi^2}{2\pi} |\sum w_s \kappa_s| = 0.5$. The peak $s(0) = 1$ occurs where $f = 1$; in decoherent regions ($f \rightarrow 0$) $s \rightarrow s_\infty = 1/(1 + \beta) = 0.67$ (weaker gravity). *Right:* SIS lensing example with a homogeneous $s = 0.90$. Convergence scales linearly, $\kappa(\theta) = s \kappa^{(0)}(\theta)$, and the Einstein radius scales as $\theta_E = s \theta_E^{(0)}$. (For a point mass, $\theta_E = \sqrt{s} \theta_E^{(0)}$.)

Coupling map and weak-field scalings. Using the one-loop master identity and coherent-domain matching, the local Newton coupling is

$$s(x) \equiv \frac{G_{\text{eff}}(x)}{G_0} = \frac{1}{1 + \frac{G_0 \Lambda_\Phi^2}{2\pi} \left(\sum_s w_s \kappa_s \right) [f(\bar{\Phi}(x)) - 1]},$$

which for the baseline content (complex scalar, massless Abelian, massive Abelian with Stückelberg) becomes

$$s(x) = \frac{1}{1 - \beta [1 - f(x)]}, \quad \beta \equiv -\frac{G_0 \Lambda_\Phi^2}{2\pi} \left(-\frac{5}{6} - 2\xi \right) > 0 \quad (\xi \geq 0).$$

As coherence decreases ($f \downarrow$), one has $s \leq 1$ and gravity weakens. In the thin-lens, single-factor regime the weak-field observables scale as $\{\kappa, \gamma\} \rightarrow s \{\kappa, \gamma\}^{(0)}$, $\Delta t \rightarrow s \Delta t^{(0)}$, and for the Einstein radius $\theta_E \rightarrow \sqrt{s} \theta_E^{(0)}$ for a point mass while $\theta_E \rightarrow s \theta_E^{(0)}$ for an SIS.

12.1 Automated checks and acceptance tests

We implemented two minimal test suites. They verify:

One-loop a_1 coefficients (species nets). Massless Abelian: $\kappa_X = -\frac{2}{3}$; Massive Abelian (Proca+Stückelberg): $\kappa_Z = -\frac{1}{2}$. For the baseline content (Φ, X_μ, Z_μ) , $\sum_s w_s \kappa_s = -\xi - 1$; checks at $\xi = 0$ (minimal) and $\xi = \frac{1}{6}$ (conformal) pass.

Coherence limits and matching consistency. With coherence-gated matching and reference = target content: $f \rightarrow 1 \Rightarrow s(x) = G_{\text{eff}}/G_0 \rightarrow 1$; $f \rightarrow 0 \Rightarrow C_R^{\text{ren}} \rightarrow 0$ and $G_{\text{eff}} \rightarrow \infty$ (EH term off); $s(x)$ is independent of Λ_Φ .

These checks close the derivation loop from the analytic Seeley–DeWitt blocks to the local coupling $s(x)$ and to the weak-field closures used in Fig. 1.

12.2 Interpretation

Using the matched expression (67) (equivalently (37)) with the Appendix C species sum $S = \sum_s w_s \kappa_s = -\frac{5}{6} - 2\xi$, we define $\beta \equiv -\frac{G_0 \Lambda_\Phi^2}{2\pi} S > 0$ for $\xi \geq 0$. Then

$$s(x) = \frac{1}{1 - \beta [1 - f(x)]}, \quad s(f=1) = 1, \quad s(f \rightarrow 0) = \frac{1}{1 + \beta} < 1.$$

Gravity gets weaker as coherence decreases. Since $1 - f \geq 0$ when coherence degrades, the denominator increases and $s \leq 1$; thus G_{eff} is reduced away from coherent domains. This trend originates in the sign of S : the Abelian gauge multiplets contribute net negative curvature-linear a_1 coefficients (massless $U(1)$: $\kappa_X^{\text{net}} = -2/3$; massive $U(1)$ in Stückelberg gauge: $\kappa_Z^{\text{net}} = -1/2$), which dominate over the scalar’s $\kappa_{\text{scalar}} = 1/6 - \xi$ so that the combined S is negative for $\xi \geq 0$.

Sign plausibility. In $d = 4$ a one-form block has $\kappa_{\text{vec}} = -1/3$; including FP ghosts (and the Stückelberg scalar in the massive case) yields the negative *net* values above. With a complex scalar ($\kappa_\Phi = 2(1/6 - \xi) = 1/3 - 2\xi$), the sum remains negative for $\xi \geq 0$, in agreement with the heat-kernel tables used in Appendix C.

How to obtain $s > 1$ if desired. Achieving $s > 1$ as f drops would require flipping S to positive, e.g. by adding sufficiently many minimally coupled scalars or by taking ξ sufficiently negative (roughly $\xi < -5/12$ so that $1/3 - 2\xi > 7/6$), subject to the usual stability and phenomenology constraints.

Relation to lensing trends. The s -factor provides a universal weak-field rescaling, $\{\kappa, \gamma\} \rightarrow s\{\kappa, \gamma\}^{(0)}$, and $\theta_E \rightarrow \sqrt{s} \theta_E^{(0)}$ for a point mass or $\theta_E \rightarrow s \theta_E^{(0)}$ for an SIS. This rescaling alone does not fix the *net* lensing strength in all environments: additional vector–sector physics (e.g. unbound Z_μ content) can enhance or suppress the potential independently of s . Hence s is a necessary but not sufficient ingredient in the full Axis-Model lens phenomenology.

13 Predictions $\rightarrow$ Falsifiability

13.1 Single-factor weak-field closures

In the coherent, low-gradient regime, adopt the operational closure

$$G_{\text{eff}}(x) = s(x) G_0, \quad s(x) \in (0, 1], \quad (68)$$

which restates Eq. (9) of Sec. 9. The Newtonian limit and lensing potential then satisfy

$$\nabla^2 \Phi_N = 4\pi G_{\text{eff}} \rho, \quad (69)$$

a restatement of Eq. (44).

Thin-lens scalings. With G_{eff} slowly varying on the lens scale,

$$\Sigma_{\text{crit}} = \frac{1}{4\pi G_{\text{eff}}} \frac{D_s}{D_l D_{ls}}, \quad (70)$$

$$\kappa(\boldsymbol{\theta}) = s \kappa^{(0)}(\boldsymbol{\theta}), \quad \gamma(\boldsymbol{\theta}) = s \gamma^{(0)}(\boldsymbol{\theta}), \quad (71)$$

$$\theta_E = \sqrt{s} \theta_E^{(0)}. \quad (72)$$

These restate Eqs. (45)–(48) from §9.

Time delays. For images A, B with Fermat potentials $\varphi(\theta, \beta)$,

$$\Delta t_{AB} = s \Delta t_{AB}^{(0)}, \quad (73)$$

recasting Eq. (50). Equivalently, Shapiro delays scale as $\Delta t_{\text{Shapiro}} \propto G_{\text{eff}}$.

Gravitational-wave strain (single-factor convention). In the same closure,

$$h(f) = s h^{(0)}(f), \quad (74)$$

and if G_{eff} varies appreciably along the path one may quote the luminosity-distance ratio

$$\ln\left(\frac{D_L^{\text{GW}}}{D_L^{\text{EM}}}\right) = \frac{1}{2} \int_0^z d\tilde{z} \frac{\partial_{\tilde{z}} \ln G_{\text{eff}}(\tilde{z})}{1 + \tilde{z}}, \quad (75)$$

which restates Eq. (53) (optional).

13.2 Concrete falsification tests

1. **Strong lenses (Einstein ring & time-delay scaling vs. environment).** At fixed mass model and geometry, test $\theta_E/\theta_E^{(0)} = \sqrt{s}$ and $\Delta t/\Delta t^{(0)} = s$ across lines of sight with different coherence proxies (e.g. LSS density, tidal field). Use a single nuisance s per LOS; seek a correlation $s(\text{environment})$ via Eq. (72) and Eq. (73).
2. **Cluster weak lensing (coherent s and substructure).** Fit a single s to the azimuthal profiles and search for coherent residuals $\delta\kappa/\kappa^{(0)} \simeq \delta\gamma/\gamma^{(0)} \simeq s(\theta) - \bar{s}$ using Eqs. (71).
3. **Standard sirens with EM counterparts.** Under Eq. (74), $D_L^{\text{EM}}/D_{L,\text{GR}}^{\text{GW}} \simeq s$; with redshift evolution use Eq. (75). Compare events that traverse distinct large-scale environments.
4. **Laboratory & solar-system nulls.** In uniform-coherence domains, Eqs. (69)–(72) reduce to GR scalings ($s \rightarrow 1$); no deviations are expected.

13.3 Scope and non-claims

We do not model Planck-scale spacetime, singularities, the problem of time, or UV completion here. The present paper treats the semiclassical, low-curvature regime; UV evolution and two-loop matching are deferred to the companion EFT/UV analyses.

14 Discussion

Our matched one-loop expression

$$s(x) \equiv \frac{G_{\text{eff}}(x)}{G_0} = \frac{1}{1 + \frac{G_0 \Lambda_\Phi^2}{2\pi} S [f(\bar{\Phi}(x)) - 1]} = \frac{1}{1 + \beta [1 - f(x)]}, \quad \beta \equiv -\frac{G_0 \Lambda_\Phi^2}{2\pi} S, \quad (76)$$

quantifies how scalar coherence modulates the local Newton coupling through a single scalar weight $f(\bar{\Phi})$ multiplying the projected trace. The species sum $S = \sum_s w_s \kappa_s$ encodes microscopic content via the heat-kernel a_1 coefficients; the projector enters only as a weight in the trace and hence does not alter block definitions. In the baseline content $S < 0$ for $\xi \geq 0$, so $\beta \geq 0$ and the map is monotone: $f \downarrow \Rightarrow s \downarrow \leq 1$, with $s(1) = 1$ and $s(0) = 1/(1 + \beta)$. This cleanly separates (i) *how coherent the domain is* (through f) from (ii) *what the field content is* (through S).

Significance within the Axis Model. Equation (76) operationalizes the Axis-Model claim that GR is the low-energy limit of a scalar-coherent sector: the EH term is the leading curvature operator and its coefficient is calculable from microphysics and coherence. The dependence is local (no nonlocal memory) and gauge/BRST-insensitive at one loop, since the ξ -dependence cancels within the projected multiplets after ghosts/Stückelberg completion. This anchors the program’s central object $G_{\text{eff}}(x)$ in standard EFT/heat-kernel machinery while keeping the projector’s role explicit and minimal.

Assumptions and limitations. Our results rest on three pillars: (i) EH emergence in coherent domains, established via the conditional path integral: the EH term appears as the curvature-linear piece in the projected one-loop action; incoherent regions do not contribute curvature. (We do not re-derive emergence here; we match its coefficient.) (ii) One-loop truncation in a low-curvature window, so that a_1 fixes the R coefficient and higher-curvature pieces enter as $\mathcal{O}(R^2, \nabla R)$. (iii) Slow gradients of the prefactor $M^2(x) = 1/(8\pi G_{\text{eff}}(x))$ on the scales probed, so the principal symbol is GR-like and gradient mixings are $\mathcal{O}(\epsilon_{\text{grad}})$. Beyond this window, higher loops renormalize the EH coefficient and populate R^2 -type operators; they do not remove the EH structure. We do not re-derive the existence of the EH operator; we compute its local coefficient in coherent domains and show that GR ($G_{\text{eff}} = G_0$) is recovered pointwise in the uniform-coherence limit. The result is an EFT statement about the EH prefactor, not a UV-completeness claim.

Connection to phenomenology. For weak-field, thin-lens and slowly varying G_{eff} , the single-factor closure $G_{\text{eff}} = sG_0$ yields the operational rules

$$\{\kappa, \gamma\} \rightarrow s \{\kappa, \gamma\}^{(0)}, \quad \theta_E \rightarrow \begin{cases} \sqrt{s} \theta_E^{(0)}, & \text{point mass,} \\ s \theta_E^{(0)}, & \text{SIS,} \end{cases} \quad \Delta t \rightarrow s \Delta t^{(0)}, \quad h \rightarrow s h^{(0)},$$

and the catalog-level invariant $R_{\Delta t} = (R_\theta)^2$, which is difficult for common lensing degeneracies to mimic jointly. When spatial variation matters, use pixelwise scaling in lenses and the standard “effective friction” relation for sirens to connect $G_{\text{eff}}(z)$ to $D_L^{\text{GW}}/D_L^{\text{EM}}$. The surrogate $s(\alpha, \zeta) = \alpha^2 e^{-\zeta}$ provides a compact, empirical parameterization (coherence amplitude α , decoherence depth ζ) consistent with the single-factor regime.

Interpretation of the sign and limits. With $\beta \geq 0$ (baseline content), $s \in [1/(1 + \beta), 1]$ and increases with f . This dovetails with the coherence limits: $f \rightarrow 1 \Rightarrow G_{\text{eff}} \rightarrow G_0$ (GR recovered), while in incoherent regions the projected trace vanishes and the EH term switches off. The phenomenological message is transparent: coherence strengthens gravity up to its GR value; loss of coherence weakens it in a controlled, bounded way.

Incoherent interiors and information. When coherence fails ($f \rightarrow 0$) the projected one-loop trace vanishes and the EH term switches off; there are no propagating spin-2 modes (Appendix B.8). This work focuses on the coherent domain and its interface, rather than attempting a unitarity proof for fully incoherent regions. A boundary-EFT description (membrane-style matching of conserved fluxes) is the natural tool for information bookkeeping across a decoherence shell and lies with the UV completion; the present results specify the coherent-side data unambiguously.

Outlook and open calculations. Two immediate directions follow: (i) Explicit S (or C_s) evaluation: complete the species-sum tables for the baseline and variants, including regulator/matching bookkeeping (proper-time vs. dim. reg.) and publish a machine-readable dictionary (the computational recipe is already laid out). (ii) Beyond one loop: implement two-loop RG evolution of the EH coefficient and quantify the leading $R^2, \nabla R$ truncation error as a systematic in the data fits—controlled within the EFT window—without altering the existence or sign of the EH term. On the data side, the pre-registered tests (Einstein angle, time delays, sirens) and the deterministic artifact suite (images, checksums, code hash) lock falsifiability and enable rapid cross-checks across lensing and GW catalogs.

Related consistency checks. The linearized equations with variable G_{eff} maintain $c_T=1$ in coherent limits, and the background-field/BRST construction ensures gauge insensitivity of the curvature-linear coefficient used throughout—points relevant for precision constraints and interpretation of null tests.

15 Conclusion

We have derived a *closed-form, local* expression for the emergent Newton coupling in scalar-coherent domains and established its formal properties:

$$s(x) = \frac{G_{\text{eff}}(x)}{G_0} = \frac{1}{1 + \beta [1 - f(x)]} \ , \quad \beta \equiv -\frac{G_0 \Lambda_\Phi^2}{2\pi} S$$

with $S = \sum_s w_s \kappa_s$ fixed by field content. The map is reparametrization- and gauge/BRST-invariant at one loop, bounded and Lipschitz on the coherent domain, and it reduces weak-field observables to a single-factor scaling s that is directly testable by joint lensing and siren data. Together with our pre-registered protocols and reproducible artifacts, this places the Axis-Model emergent gravity program on a falsifiable, computationally transparent footing.

Reproducibility.

All figures can be regenerated directly from the **Python/ Jupyter** reproducibility suite released with this paper (available under the shared Zenodo DOI: <https://doi.org/10.5281/zenodo.17449079>). Researchers can reproduce, extend, or cross-check all results by cloning the Zenodo repository and executing the supplied notebooks.

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References

- [1] A. D. Sakharov. Vacuum quantum fluctuations in curved space and the theory of gravitation. *Sov. Phys. Dokl.*, 12:1040–1041, 1968. Dokl. Akad. Nauk SSSR 177 (1967) 70–71.
- [2] Ted Jacobson. Thermodynamics of spacetime: The einstein equation of state. *Phys. Rev. Lett.*, 75:1260–1263, 1995.
- [3] John F. Donoghue. General relativity as an effective field theory: The leading quantum corrections. *Phys. Rev. D*, 50:3874–3888, 1994.
- [4] A. O. Barvinsky and G. A. Vilkovisky. The generalized schwinger–dewitt technique in gauge theories and quantum gravity. *Phys. Rep.*, 119:1–74, 1985.
- [5] Dmitri V. Vassilevich. Heat kernel expansion: User’s manual. *Phys. Rep.*, 388:279–360, 2003.
- [6] Peter Schneider, Christopher S. Kochanek, and Joachim Wambsganss. *Gravitational Lensing: Strong, Weak and Micro*. Saas-Fee Advanced Course 33. Springer, Berlin, Heidelberg, 2006.
- [7] E. E. Falco, M. V. Gorenstein, and I. I. Shapiro. On model-dependent bounds on h_0 from gravitational lenses. *Astrophys. J. Lett.*, 289:L1–L4, 1985.
- [8] Tommaso Treu and Philip J. Marshall. Time delay cosmography. *Astron. Astrophys. Rev.*, 24(11), 2016.
- [9] Bernard F. Schutz. Determining the hubble constant from gravitational wave observations. *Nature*, 323:310–311, 1986.
- [10] LIGO Scientific Collaboration and Virgo Collaboration. A gravitational-wave standard siren measurement of the hubble constant. *Nature*, 551:85–88, 2017.
- [11] Andrew Morton. Quantum gravitational extension of the axis model, 2025.
- [12] Andrew Morton. The axis model: A unified framework for emergent particle structure, cosmology, and gravitational phenomena. Zenodo, 2025. Comprehensive overview of the Axis Model framework and emergent gravitational coupling.
- [13] Andrew Morton. Quantum completion of the axis model: Gauge structure, brst invariance, and renormalization stability. Zenodo preprint, 2025. Sections cited: §§1.3, 2.2–2.3.
- [14] Andrew Morton. Quantum consistency and renormalization of the axis model effective field theory. Preprint; reproducibility suite: <https://doi.org/10.5281/zenodo.17370236>, October 2025. Sections cited: §3.7 and Appendix A.4.

A Curvature-Linear Extraction and Non- R Pieces

Consider a minimal Laplace-type operator $D_s = -\nabla^2 \mathbf{1} + E_s$ in $d = 4$. The local heat-kernel coefficient is

$$a_1^{(s)}(x) = \text{tr}\left(\frac{1}{6}R\mathbf{1} - E_s\right).$$

Decompose $E_s = E_s^{(R)} + E_s^{(-R)}$, where $E_s^{(R)}$ is linear in the curvature (linear combination of R and $R^\mu{}_\nu$ after bundle trace) and $E_s^{(-R)}$ contains masses, potentials, and composite-geometry terms with no curvature. Then

$$a_{1,R}^{(s)}(x) = \text{tr}\left(\frac{1}{6}\mathbf{1} - E_s^{(R)}\right) R(x),$$

while $E_s^{(-R)}$ contributes only to a_0 and higher non-geometric local terms (cosmological constant, wavefunction renormalizations, etc.). Hence the induced Einstein–Hilbert coefficient depends only on $E_s^{(R)}$, justifying the use of κ_s as the R -coefficient.

B One-loop Einstein–Hilbert coefficient (proved)

Statement. In the scalar-coherent domain and to linear order in curvature, the conditional / projected path integral for the Axis Model yields a local, covariant Einstein–Hilbert (EH) term in the effective action,

$$S_{\text{eff}}[g] \supset \frac{1}{16\pi G_{\text{eff}}(x)} \int d^4x \sqrt{-g} R + \mathcal{O}(R^2, \nabla R, \dots), \quad (77)$$

with $G_{\text{eff}}(x)$ determined by the one-loop Seeley–DeWitt a_1 coefficients of the projected fluctuation operators and a standard finite counterterm fixing scheme. (Emergence of the EH *structure* from the conditional / filtered path integral is reviewed in the main text and in the Quantum Gravitational Extension.)

B.1 Setup: conditional / projected path integral and background field method

We define the conditional effective action by coarse-graining over internal microstates at fixed emergent metric $g_{\mu\nu}$:

$$e^{iS_{\text{eff}}[g]} = \int \mathcal{D}v \mathcal{D}\Phi \delta[g_{\mu\nu} - g_{\mu\nu}[v, \Phi]] \Pi_\Phi[v, \Phi] e^{iS[v, \Phi]}, \quad (78)$$

with $g_{\mu\nu}[v, \Phi]$ the composite metric built from the emergent frame and Π_Φ the scalar-coherence projector. In practice we employ the background-field method: expand about a coherent background $(\bar{g}, \bar{\Phi}, \bar{v})$ that reproduces g on shell and integrate the projected fluctuations.

The projection weight enters the one-loop trace as a *scalar* multiplicative factor $f(\bar{\Phi}(x)) \in (0, 1]$: $\text{Tr}_\Phi(\mathcal{O}) = \int \sqrt{-\bar{g}} f(\bar{\Phi}) \text{tr}\langle x | \mathcal{O} | x \rangle$. This preserves covariance at this order (see B.4).

B.2 Reduction to Laplace type and the projected heat kernel

After covariant gauge-fixing of the projected vector sectors and inclusion of their ghosts (and Stückelberg scalars for massive vectors), local field redefinitions bring each quadratic block to minimal Laplace type $D_s = -\nabla^2 \mathbf{1} + E_s$, with s running over the projected physical and ghost blocks. The one-loop piece is

$$S^{(1)}[g; \bar{\Phi}] = \frac{i}{2} \sum_s w_s \text{Tr}_\Phi \ln D_s = -\frac{i}{2} \sum_s w_s \int_{\Lambda_\Phi^{-2}}^\infty \frac{dt}{t} \text{Tr}_\Phi e^{-tD_s}. \quad (79)$$

Expanding the (projected) heat kernel in $d=4$: $\text{Tr}_\Phi e^{-tD_s} = (4\pi t)^{-2} \int \sqrt{-g} f(\bar{\Phi}) [a_0^{(s)} + t a_1^{(s)} + \mathcal{O}(t^2)]$, with $a_1^{(s)} = \text{tr}(\frac{1}{6} R \mathbf{1} - E_s)$. Only $a_1^{(s)}$ contributes to the curvature-linear term.

For each block define $a_{1,R}^{(s)}(x) \equiv \kappa_s(x) R(x)$, so that for a real scalar with nonminimal coupling ξ , $\kappa_{\text{scalar}} = \frac{1}{6} - \xi$; for a vector on one-forms, $E^\mu{}_\nu = R^\mu{}_\nu + \dots \Rightarrow \kappa_{\text{vec}} = \frac{d}{6} - 1 = -\frac{1}{3}$ in $d=4$. Ghosts and Stückelberg scalars contribute as minimally coupled scalars with $\kappa = \frac{1}{6}$, with statistics / multiplicity in w_s . index=4

B.3 Matching to the EH structure and the master identity

Performing the proper-time integral gives the curvature-linear, *local* one-loop piece

$$S_R^{(1)} = \frac{1}{2} \sum_s w_s \frac{\Lambda_\Phi^2}{(4\pi)^2} \int d^4x \sqrt{-g} f(\bar{\Phi}) \kappa_s(x) R(x) + (\text{scheme-dependent local terms}). \quad (80)$$

Matching (80) to the EH form yields the local master identity

$$\boxed{\frac{1}{16\pi G_{\text{eff}}(x)} = \frac{\Lambda_\Phi^2}{(4\pi)^2} \frac{1}{2} f(\bar{\Phi}(x)) \sum_s w_s \kappa_s(x) + \delta\left(\frac{1}{16\pi G}\right)_{\text{ren}}(x)}, \quad (81)$$

where the δ counterterm fixes the finite piece in the chosen renormalization scheme. Choosing a coherent reference $(*)$ with $f(\bar{\Phi}_*) \simeq 1$ and imposing $G_{\text{eff}}|_* = G_0$ fixes the finite part and gives the usual matched formulae for the local ratio $s(x) \equiv G_{\text{eff}}(x)/G_0$.

Species nets (for immediate use). Massless Abelian X_μ (vector+ghost): $\kappa^{\text{net}} = -\frac{2}{3}$. Massive Abelian Z_μ (vector+Stückelberg+ghost): $\kappa^{\text{net}} = -\frac{1}{2}$. Hence for the baseline content (Φ, X_μ, Z_μ) , $\sum_s w_s \kappa_s = (\frac{1}{6} - \xi) - \frac{2}{3} - \frac{1}{2} = -\xi - 1$, as used in the main text's matched expressions.

B.4 Rigor checks at this order

(i) Locality to $\mathcal{O}(R)$. The only $\mathcal{O}(R)$ contribution arises from the a_1 coefficient in the heat-kernel, hence is pointwise local; nonlocalities first enter at $\mathcal{O}(t^2) \sim \mathcal{O}(R^2, \nabla R)$. Equation (80) is thus a local density. Boundary terms (e.g. GHY) lie outside the bulk one-loop analysis.

(ii) Covariance preserved by the projection. We work in background-field gauge; heat-kernel coefficients are built from covariant tensors $(R, R^\mu{}_\nu, \dots)$. The projector contributes only the scalar weight $f(\bar{\Phi}(x))$ inside the trace, which does not break diffeomorphism invariance. For gauge sectors, BRST ensures that the gauge-parameter dependence cancels between vectors, ghosts, and (where applicable) Stückelberg fields in the local a_1 . Therefore the $\mathcal{O}(R)$ coefficient is gauge- and diffeo-invariant at one loop.

(iii) Regulator / scheme independence of the curvature-linear coefficient. The coefficient extracted from a_1 (i.e. the κ_s and their weighted sum) is universal; differences among proper-time cutoffs and dimensional regularization appear only as local additions that are absorbed into $\delta(1/16\pi G)_{\text{ren}}$. In dim. reg. the same a_1 controls the $1/\epsilon$ pole; the matched G_{eff} is scheme-independent once the finite counterterm is fixed at the reference background.

Combining (i)–(iii) with (81) proves (77).

B.5 Independent cross-check (Schwinger–DeWitt / diagrammatics) on a toy background

Toy background. Take a slowly varying Einstein background with constant scalar curvature R (e.g. S^4 or Euclidean dS_4) or, equivalently, a conformally flat metric $g_{\mu\nu} = e^{2\sigma(x)}\eta_{\mu\nu}$ with $|\partial\sigma| \ll \Lambda_\Phi$. For a minimally coupled real scalar,

$$\ln \det(-\nabla^2 + \xi R) \Rightarrow \frac{\Lambda_\Phi^2}{(4\pi)^2} \int \sqrt{-g} \left(\frac{1}{6} - \xi \right) R \quad \text{at } \mathcal{O}(R),$$

reproducing $\kappa_{\text{scalar}} = \frac{1}{6} - \xi$. For a massless Abelian vector, Lichnerowicz form gives $E^\mu{}_\nu = R^\mu{}_\nu$ so $\kappa_{\text{vec}} = -\frac{1}{3}$; adding the complex FP ghost (two real scalars with Grassmann weight) yields $\kappa_X^{\text{net}} = -\frac{2}{3}$. For a massive Abelian in Stückelberg gauge, including χ and the massive ghost gives $\kappa_Z^{\text{net}} = -\frac{1}{2}$. Summing the baseline content reproduces the analytic net $\sum_s w_s \kappa_s = -\xi - 1$ used in the master identity.

Diagrammatic (two-point) check. Expand $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$ in Riemann normal coordinates about a point on the toy background and compute the one-loop 1PI two-point function for $h_{\mu\nu}$ with the same projected field content and BRST-consistent gauges. The local, momentum-space piece $\propto q^2 h h$ that survives as $q \rightarrow 0$ matches the linearized EH kernel and carries the same κ -sums as above; by covariant completion, this reproduces the $\int \sqrt{-g} R$ term with exactly the coefficients extracted from a_1 . This verifies the a_1 result by an independent, explicitly diagrammatic route (the “Schwinger–DeWitt vs. 2-point” cross-check). (See the BRST capsule and gauge-fixing structure in the companion quantum-completion notes.)

B.6 Notes linking back to the main construction

- (1) The conditional / filtered coarse-graining that produces the EH lead term is developed in Sec. 3.1, which defines the conditional path integral and the role of scalar coherence.
- (2) The precise placement of the projection weight $f(\bar{\Phi})$ in the trace (as a scalar factor) and the master identity (81) match the derivation in the gravity note; quick-use tables for κ_s and the net gauge multiplets are provided there for reproducibility.

B.7 Lemma (Positivity of the EH coefficient in coherent domains).

Let $f(\bar{\Phi})$ be the scalar-coherence weight and $S \equiv \sum_s w_s \kappa_s$ the one-loop R -coefficient sum for the projected fluctuation blocks. With the coherent-domain matching condition $G_{\text{eff}} = G_0$ at $f \simeq 1$, the master identity gives

$$\frac{1}{16\pi G_{\text{eff}}(x)} = \frac{1}{16\pi G_0} + \frac{\Lambda_\Phi^2}{(4\pi)^2} \frac{1}{2} [f(\bar{\Phi}(x)) - 1] S. \quad (82)$$

In every *coherent* region one has $0 < f(\bar{\Phi}) \leq 1$, and for the Axis-baseline content (Φ, X_μ, Z_μ) with nonminimal scalar coupling $\xi \geq 0$,

$$S = \left(\frac{1}{6} - \xi \right) + \left(-\frac{2}{3} \right) + \left(-\frac{1}{2} \right) = -\frac{5}{6} - 2\xi < 0. \quad (83)$$

Therefore $f - 1 \leq 0$ and $S < 0$ imply the correction term in (82) is *nonnegative*. Hence

$$\frac{1}{16\pi G_{\text{eff}}(x)} \geq \frac{1}{16\pi G_0} > 0 \quad \text{for } 0 < f(\bar{\Phi}) \leq 1,$$

with equality only at $f = 1$. In particular, the EH coefficient cannot flip sign within the coherent domain: the graviton kinetic term retains the healthy (non-ghost) sign. $\square$

B.8 Decoupling and GR Limit

Statement. As scalar coherence becomes uniform ($f(\bar{\Phi}) \rightarrow 1$ and $\nabla_\mu \bar{\Phi} \rightarrow 0$), the Axis-Model effective theory reduces continuously to General Relativity:

$$G_{\text{eff}} \rightarrow G_0, \quad \text{PPN} \equiv \text{GR}, \quad c_T = 1.$$

We now show this at the level of the equations of motion, not just parametrically.

Effective action and variation. From the matched one-loop result,

$$S_{\text{eff}}[g, \Phi] = \frac{1}{16\pi G_0} \int d^4x \sqrt{-g} \left[f(\Phi) R + \frac{2\pi S G_0 \Lambda_\Phi^2}{(4\pi)^2} (f(\Phi) - 1) R \right] + S_{\text{m}}[g, \psi], \quad (84)$$

or equivalently, using the master identity (Eq. (68)),

$$S_{\text{eff}} = \frac{1}{16\pi G_{\text{eff}}(\Phi)} \int d^4x \sqrt{-g} R + S_{\text{m}}[g, \psi].$$

Varying (84) with respect to the metric gives the field equations

$$f(\Phi) G_{\mu\nu} = 8\pi G_0 T_{\mu\nu} + (\nabla_\mu \nabla_\nu - g_{\mu\nu} \square) f(\Phi) - \frac{2\pi S G_0 \Lambda_\Phi^2}{(4\pi)^2} (\nabla_\mu \nabla_\nu - g_{\mu\nu} \square) (f(\Phi) - 1), \quad (85)$$

where $T_{\mu\nu}$ is the matter stress-energy tensor and we used $\delta(\sqrt{-g}R) = \sqrt{-g} (G_{\mu\nu} \delta g^{\mu\nu} + \nabla_\mu \nabla_\nu - g_{\mu\nu} \square) \delta g^{\mu\nu}$.

Uniform-coherence (GR) limit. In the limit

$$f(\Phi) \rightarrow 1, \quad \nabla_\mu f(\Phi) \rightarrow 0,$$

all derivative terms in (85) vanish and the curvature coupling becomes constant:

$$G_{\mu\nu} = 8\pi G_0 T_{\mu\nu},$$

which are precisely the Einstein field equations with Newton coupling G_0 . Hence the full GR dynamics are recovered pointwise in any scalar-homogeneous region.

Weak-field and PPN limit. Expand $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$, $f(\Phi) = 1 + \delta f$ with $|\delta f| \ll 1$. Linearizing (85) gives

$$\square \bar{h}_{\mu\nu} = -16\pi G_0 T_{\mu\nu} + \mathcal{O}(\partial^2 \delta f),$$

where $\bar{h}_{\mu\nu} = h_{\mu\nu} - \frac{1}{2} \eta_{\mu\nu} h$ in harmonic gauge. In the uniform-coherence limit $\partial^2 \delta f \rightarrow 0$, so the standard linearized Einstein equations are recovered, with the same gravitational potentials and PPN parameters:

$$\gamma_{\text{PPN}} = \beta_{\text{PPN}} = 1.$$

All post-Newtonian observables (Shapiro delay, light deflection, perihelion precession) therefore coincide with GR at leading order.

Tensor-mode propagation. The quadratic expansion of (84) in $h_{\mu\nu}$ yields the kinetic term

$$\frac{1}{64\pi G_{\text{eff}}} \int d^4x (\partial_\sigma h_{\mu\nu})^2,$$

so the graviton phase speed is

$$c_T^2 = \frac{\text{coefficient of } (\partial_t h)^2}{\text{coefficient of } (\nabla h)^2} = 1,$$

independent of $f(\Phi)$ when $\nabla_\mu f = 0$. Thus the propagation of gravitational waves is luminal in the coherent (GR) limit.

Conclusion. When the scalar field becomes spatially uniform,

$$f(\Phi) \rightarrow 1, \quad \nabla_\mu f \rightarrow 0,$$

the additional scalar-geometric terms in the effective equations decouple, leaving standard Einstein dynamics with $G_{\text{eff}} = G_0$, PPN parameters identical to GR, and $c_T = 1$. Hence the Axis-Model effective gravity is fully consistent with GR in the homogeneous-coherence limit. $\square$

C Linearized Field Equations and Stability in Variable Coupling Domains

For the benchmark sum, we treat each complex scalar as two real degrees of freedom, so that the field content used here (Φ complex, X_μ, Z_μ real vectors) corresponds to four bosonic fluctuation blocks in total.

C.1 Linearized equations with variable coupling (explicit gauge fixing)

Setup and notation. Write the Einstein–Hilbert prefactor as a position–dependent Planck mass

$$M^2(x) \equiv \frac{1}{8\pi G_{\text{eff}}(x)}, \quad S_{\text{eff}}[g, \Phi] = \frac{1}{2} \int d^4x \sqrt{-g} M^2(x) R + S_m[g, \psi], \quad (86)$$

the matched one–loop effective action in the scalar–coherent domain (master identity and domain assumptions summarized earlier).

Exact field equations. Varying (86) with respect to $g^{\mu\nu}$ yields the standard $F(x)R$ equations with $F=M^2$:

$$M^2 G_{\mu\nu} = T_{\mu\nu} + \nabla_\mu \nabla_\nu M^2 - g_{\mu\nu} \square M^2, \quad (87)$$

where $T_{\mu\nu}$ is the physical (minimally coupled) stress–energy tensor. Background-field and BRST consistency are the same ones used in the one–loop derivation of the EH coefficient.

Linearization and gauge fixing. Expand about a flat coherent background

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}, \quad M^2(x) = M_0^2 + \delta M^2(x), \quad M_0^2 = \frac{1}{8\pi G_0}.$$

Define the trace–reversed field $\bar{h}_{\mu\nu} \equiv h_{\mu\nu} - \frac{1}{2}\eta_{\mu\nu}h$ and the de Donder gauge function $C_\nu \equiv \partial^\mu \bar{h}_{\mu\nu}$. Gauge–fix with

$$\mathcal{L}_{\text{GF}} = \frac{M_0^2}{4} C_\mu C^\mu, \quad (88)$$

which is the standard harmonic gauge at the background value M_0^2 (no singular behavior in coherent domains).

Using (87) to first order in perturbations and collecting the linear pieces from the covariant derivatives acting on M^2 , one finds

$$\begin{aligned} \square \bar{h}_{\mu\nu} = & -16\pi G_0 T_{\mu\nu} + \frac{2}{M_0^2} (\partial_\mu \partial_\nu - \eta_{\mu\nu} \square) \delta M^2 \\ & + \frac{1}{M_0^2} (\partial_\lambda M^2) \left(\partial^\lambda \bar{h}_{\mu\nu} - \partial_{(\mu} \bar{h}_{\nu)}^\lambda + \eta_{\mu\nu} C^\lambda \right) + \underbrace{\left(\partial_\mu C_\nu + \partial_\nu C_\mu - \eta_{\mu\nu} \partial \cdot C \right)}_{\text{vanishes in harmonic gauge}}, \end{aligned} \quad (89)$$

where we used $G_{\mu\nu}^{(1)} = -\frac{1}{2}\square \bar{h}_{\mu\nu} + \partial_{(\mu} C_{\nu)} - \frac{1}{2}\eta_{\mu\nu} \partial \cdot C$.

Weak-gradient (closure) power counting. For modes with $(\omega, \mathbf{k})$, the GR term scales as $\omega^2 \bar{h}$, while the new pieces scale like

$$\frac{|\partial M^2|}{M_0^2} |\mathbf{k}| |\bar{h}|, \quad \frac{|\partial^2 M^2|}{M_0^2} |\bar{h}|.$$

Define the small parameter

$$\epsilon \equiv \max \left\{ \frac{|\partial M^2|}{M_0^2 |\mathbf{k}|}, \frac{|\partial^2 M^2|}{M_0^2 \omega^2} \right\} \ll 1. \quad (90)$$

Then the extra terms in (89) are $\mathcal{O}(\epsilon)$ and do not change the weak-field “single-factor” rules (replace $G_0 \mapsto G_{\text{eff}}$ in sources; same kernels). This is the same closure used in the observable map (PPN, lensing, GW amplitudes).

No hidden longitudinal/ghost mode. The gauge condition $C_\nu = 0$ removes non-TT modes as in GR; the ∂M^2 terms carry at most one derivative of h and thus do not alter the kinetic rank or excite an extra propagating longitudinal/FP ghost mode. BRST arguments guaranteeing gauge-parameter independence of the curvature-linear coefficient at one loop are unchanged.

In harmonic gauge,

$$\square \bar{h}_{\mu\nu} = -16\pi G_{\text{eff}} T_{\mu\nu}^{\text{phys}} + (\nabla G_{\text{eff}}) \cdot [\text{gauge-fixed, lower-order terms}],$$

with the (∇G_{eff}) terms suppressed by ϵ in (90).

C.2 Hyperbolicity and Causality (principal symbol; $c_T = 1$)

Principal symbol in the coherent domain. The differential operator acting on h in (89) has principal (highest-derivative) part

$$\mathcal{P}^{\alpha\beta}{}_{\mu\nu} \partial_\alpha \partial_\beta h^{\mu\nu} = -\frac{M_0^2}{2} \square \bar{h}_{\mu\nu},$$

while the terms containing ∂M^2 multiply at most first derivatives of h (and $\partial^2 M^2$ multiplies h without derivatives). Hence the *principal symbol* is identical to GR’s:

$$\sigma(\mathcal{P}; k) \bar{h}_{\mu\nu} = -\frac{M_0^2}{2} (k \cdot k) \bar{h}_{\mu\nu},$$

so the characteristics satisfy $k \cdot k = 0$ (wave cones unchanged) and the tensor speed is luminal,

$$c_T = 1 \quad (\text{coherent domain}). \quad (91)$$

This matches the coherent-limit statement obtained from the GR-limit analysis.

Toy counterexample outside coherence (boundary marker). Let $M^2(x) = M_0^2 f(x)$ with $f(x) \rightarrow 0$ across a thin layer of width L (decoherence boundary). The principal part becomes $-\frac{1}{2} M_0^2 f(x) \square \bar{h}$, which *degenerates* as $f \rightarrow 0$ and loses hyperbolicity—precisely where the EH term switches off and spacetime dynamics dissolve in the projection framework (outside the stated domain).

C.3 No tachyon/gradient instabilities (quadratic action; eigenvalue bounds)

Quadratic action about a coherent background. Expanding the gravitational action plus (88) to quadratic order and keeping the leading mixings from ∂M^2 gives

$$S_{\text{grav+GF}}^{(2)} = \frac{M_0^2}{8} \int d^4x \left[(\partial_\lambda h_{\mu\nu})^2 - (\partial_\lambda h)^2 + 2 \partial_\mu h^{\mu\nu} \partial_\nu h - 2 \partial_\mu h^{\mu\nu} \partial^\lambda h_{\lambda\nu} \right] + \frac{M_0^2}{4} \int d^4x C_\mu C^\mu + \int d^4x \left[\frac{1}{2} (\partial_\lambda M^2) \mathcal{I}^\lambda[h] + \frac{1}{2} (\partial_\mu \partial_\nu M^2) \mathcal{Q}^{\mu\nu}[h] \right], \quad (92)$$

where $\mathcal{I}^\lambda[h]$ is linear in $h \partial h$ and $\mathcal{Q}^{\mu\nu}[h]$ is linear in h (their explicit forms follow from the same linearization that produced (89)).

In harmonic gauge $C_\mu=0$, only the transverse–traceless (TT) block is dynamical:

$$S_{\text{TT}}^{(2)} = \frac{M_0^2}{8} \int d^4x \left[(\partial_t h_{ij}^{\text{TT}})^2 - (\nabla h_{ij}^{\text{TT}})^2 \right] + \int d^4x \delta \mathcal{L}_{\text{mix}}^{\text{TT}},$$

$$|\delta \mathcal{L}_{\text{mix}}^{\text{TT}}| \lesssim \frac{|\partial M^2|}{M_0^2} |\partial h^{\text{TT}}| |h^{\text{TT}}| + \frac{|\partial^2 M^2|}{M_0^2} |h^{\text{TT}}|^2. \quad (93)$$

Eigenvalue bounds. Let Φ collect the independent TT Fourier amplitudes. The quadratic form reads

$$\delta^2 S_{\text{TT}} = \frac{1}{2} \int d^4x \Phi^\top \left[\underbrace{\mathbb{K}_0}_{\frac{M_0^2}{4} \mathbf{1} \cdot \omega^2} - \underbrace{\mathbb{G}_0}_{\frac{M_0^2}{4} \mathbf{1} \cdot \mathbf{k}^2} + \Delta(\partial M^2) \right] \Phi,$$

with $\|\Delta(\partial M^2)\| \lesssim c_1 \frac{|\partial M^2|}{M_0^2} |\mathbf{k}| + c_2 \frac{|\partial^2 M^2|}{M_0^2}$. Gershgorin/perturbation bounds imply

$$\lambda_{\text{kin}} \geq \frac{M_0^2}{4} \omega^2 (1 - \mathcal{O}(\epsilon)), \quad \lambda_{\text{grad}} \geq \frac{M_0^2}{4} \mathbf{k}^2 (1 - \mathcal{O}(\epsilon)), \quad (94)$$

where ϵ is defined in (90). Thus the TT sector has *positive* kinetic and gradient eigenvalues (no ghost/tachyon/gradient instabilities) in the coherent domain; vector/scalar pieces remain nondynamical under $C_\mu=0$. Positivity of $M^2(x)$ in coherent regions was proved in the EH-sign lemma.

In scalar-coherent regions the principal symbol and characteristics coincide with GR (hyperbolic, $c_T=1$), and the quadratic TT action has strictly positive kinetic/gradient terms; departures induced by ∇M^2 are lower-order and $\mathcal{O}(\epsilon)$ -suppressed.

D Conditional Expectation vs. Functional Delta

Starting from

$$e^{iS_{\text{eff}}[g]} = \int \mathcal{D}v \mathcal{D}\Phi \delta[g - g[v, \Phi]] \Pi_\Phi e^{iS[v, \Phi]},$$

replace the delta by either (i) a Lagrange multiplier (sharp constraint) or (ii) a narrow Gaussian (soft constraint) of width κ^{-1} . Expanding about a coherent background and keeping terms through quadratic order in fluctuations yields kernels that differ only by local contact terms (polynomials in background fields) which are absorbed into renormalized couplings (cosmological constant, R^2 terms, etc.). Thus the one-loop *quadratic* kernel and its a_1 coefficient are independent of the technical implementation of the constraint. The same reasoning applies to the operator projector: under the one-loop, background-field trace, $P_\Phi \rightarrow f(\Phi)$ acts as a scalar weight.

E Check of the species sum $\sum_s w_s \kappa_s$

Field content and conventions. We consider a complex scalar Φ with nonminimal coupling ξ , one massless Abelian gauge field X_μ (including Faddeev–Popov ghosts), and one massive Abelian gauge field Z_μ in Stückelberg gauge (vector + Stückelberg scalar χ + ghosts). Throughout, w_s denotes the statistics/multiplicity weight (+1 for bosons, -1 for fermions), with multiplicity counted per real degree of freedom. For the gauge sectors we use *net* curvature–linear coefficients κ^{net} in which ghost/Stückelberg contributions are already included; hence no separate ghost lines appear in the sum.

Curvature–linear (a_1) blocks. In $d = 4$, the Seeley–DeWitt a_1 coefficient for a Laplace–type block contributes to the Einstein–Hilbert term via

$$\frac{1}{16\pi G} \supset \frac{1}{2} \frac{\Lambda_\Phi^2}{(4\pi)^2} f(\bar{\Phi}) \sum_s w_s \kappa_s.$$

For the stated content the relevant entries are:

$$\begin{aligned} \text{(real) scalar: } \kappa_{\text{scalar}}(\xi) &= \frac{1}{6} - \xi, \\ \text{(complex) scalar: } \kappa_\Phi &= 2\left(\frac{1}{6} - \xi\right) = \frac{1}{3} - 2\xi, \\ \text{massless } U(1) \text{ (net): } \kappa_X^{\text{net}} &= -\frac{2}{3}, \\ \text{massive } U(1) \text{ (net, Stückelberg): } \kappa_Z^{\text{net}} &= -\frac{1}{2}. \end{aligned} \tag{95}$$

For quick reference:

Block	κ_s	Comment
Φ (complex scalar)	$\frac{1}{3} - 2\xi$	two real dof
X_μ (massless $U(1)$, net)	$-\frac{2}{3}$	vector + FP ghosts
Z_μ (massive $U(1)$, net)	$-\frac{1}{2}$	vector + χ + FP ghosts

Species sum. Using (95), the weighted sum entering the EH prefactor is

$$\boxed{\sum_s w_s \kappa_s = \underbrace{\left(\frac{1}{3} - 2\xi\right)}_\Phi + \underbrace{\left(-\frac{2}{3}\right)}_{X_\mu \text{ (net)}} + \underbrace{\left(-\frac{1}{2}\right)}_{Z_\mu \text{ (net)}} = -\frac{5}{6} - 2\xi} \tag{96}$$

Sanity checks.

$$\xi = 0 \Rightarrow \sum_s w_s \kappa_s = -\frac{5}{6}, \quad \xi = \frac{1}{6} \Rightarrow \sum_s w_s \kappa_s = -\frac{7}{6}.$$

The negative sign is expected: the Abelian gauge multiplets (vector + ghosts, and for the massive case also the Stückelberg scalar) yield net negative a_1 coefficients, which dominate over the scalar’s positive $(\frac{1}{6} - \xi)$ contribution.

Remark on weights. We take $w_s = +1$ for bosonic blocks; multiplicities are included explicitly (e.g., the complex scalar carries a factor of 2 from two real dof). The *net* entries κ_X^{net} and κ_Z^{net} already include the Faddeev–Popov/Stückelberg contributions with their statistics signs, so no further w_s factors are applied to ghosts separately.

F Empirical surface: falsifiability locked

Scope and principle. This appendix freezes the empirical tests that would falsify the single-factor observable map. All statistics, cuts, and decision rules are pre-declared here. The *code and deterministic artifacts* live in the companion Colab notebook (see §F.2).

F.1 Pre-registered tests (frozen statistics and cuts)

Null and alternative. Unless stated otherwise, test the null $H_0 : s = 1$ against the two-sided alternative $H_1 : s \neq 1$. For each test, compute an estimator $\hat{s}$ with variance $\sigma_{\hat{s}}^2$ from the frozen likelihoods in Sec. 11. Primary decision rule:

$$Z \equiv \frac{\hat{s} - 1}{\sigma_{\hat{s}}}, \quad \text{reject } H_0 \text{ if } |Z| \geq z_\alpha, \quad (97)$$

with $\alpha = 0.01$ (two-sided), $z_\alpha = 2.576$. Family-wise error is controlled by Bonferroni across T1–T3 (per-test $\alpha = 0.01/3$) and by FDR $q = 0.05$ for any catalog scans. A secondary joint decision uses $\delta_i \equiv R_{\Delta t, i} - (R_{\theta, i})^2$ with $\chi^2 = \sum_i \delta_i^2 / \sigma_{\delta_i}^2$ and a 1% upper-tail threshold.

Frozen data products. For each test we freeze: (i) object lists (IDs, redshifts, quality flags); (ii) per-object measurement uncertainties; (iii) baseline values $\{\theta_E^{(0)}, \Delta t^{(0)}, h^{(0)}\}$; (iv) nuisance priors used for marginalization. The frozen lists are versioned by a repository commit hash (recorded with the artifacts in §F.2).

T1 — Einstein angle test

Observable. $r_{\theta, i} \equiv \theta_{E, i} / \theta_{E, i}^{(0)} = \sqrt{s}$.

Estimator. Weighted mean of $\sqrt{s}$, $\widehat{\sqrt{s}} = (\sum_i w_i r_{\theta, i}) / (\sum_i w_i)$, $w_i = \sigma_{r_{\theta, i}}^{-2}$; report $s_{T1} = (\widehat{\sqrt{s}})^2$ with $\sigma_{s, T1}^2 = 4 \widehat{\sqrt{s}}^2 \text{Var}(\widehat{\sqrt{s}})$.

Cuts (frozen). Axisymmetric/ring lenses (morphology ≥ 3); $\sigma(\theta_E) / \theta_E \leq 3\%$; external shear $|\gamma_{\text{ext}}| \leq 0.1$; spectroscopic z_l, z_s .

Falsifier. Reject H_0 if $|Z_{T1}| \geq z_{\alpha/3}$. Release residuals $\varepsilon_{\theta, i} = r_{\theta, i} - \widehat{\sqrt{s}}$.

T2 — Time-delay test

Observable. $r_{\Delta t, i} \equiv \Delta t_i / \Delta t_i^{(0)} = s$.

Estimator. $s_{T2} = \frac{\sum_i \Delta t_i^{(0)} \Delta t_i / \sigma_{\Delta t, i}^2}{\sum_i (\Delta t_i^{(0)})^2 / \sigma_{\Delta t, i}^2}$, with $\sigma_{s, T2}^{-2} = \sum_i (\Delta t_i^{(0)})^2 / \sigma_{\Delta t, i}^2$.

Cuts (frozen). Fractional delay error $\leq 5\%$; successful source re-fit (stationary Fermat points); same shear/environment cuts as T1; no microlensing correction exceeding 2σ of macro residuals.

Falsifier. Reject H_0 if $|Z_{T2}| \geq z_{\alpha/3}$. Also test per-system $\delta_i = r_{\Delta t, i} - r_{\theta, i}^2$; significant catalog χ^2 rejects the single-factor map independent of s .

T3 — Standard-siren amplitude test

Observable. For each GW event with (M_c, D_L) and reference frequency f ,

$$r_{h, k} \equiv h_k / h_k^{(0)}(M_c, D_L, f) = s$$

(operational closure).

Estimator. Inverse-variance mean: $s_{T3} = \frac{\sum_k \hat{s}_k / \sigma_{s, k}^2}{\sum_k 1 / \sigma_{s, k}^2}$, $\sigma_{s, T3}^{-2} = \sum_k 1 / \sigma_{s, k}^2$, where $(\hat{s}_k, \sigma_{s, k})$ come from the event PE chain marginalized over calibration and inclination.

Cuts (frozen). Network SNR $\rho \geq 12$; sky area $\leq 50 \text{ deg}^2$ (90%); EM host redshift with $\sigma_z/(1+z) \leq 0.01$; calibration priors fixed to public release; discard events with $E[\mu] > 1.2$ unless corrected.

Falsifier. Reject H_0 if $|Z_{T3}| \geq z_{\alpha/3}$. Cross-check consistency with lensing by requiring $|s_{T3} - s_{T1/T2}| \leq \sqrt{\sigma_{s,T3}^2 + \sigma_{s,T1/T2}^2}$.

Unblinding protocol. (1) Freeze object lists, baselines, cuts, and code (commit hash). (2) Dry-run pipelines on *scrambled* nuisance seeds to validate tables/plots only. (3) Compute power curves for $|s - 1| \in \{0.02, 0.05, 0.10\}$ from frozen variances and N . (4) Unblind: compute (s_{T1}, s_{T2}, s_{T3}) , joint χ^2 on δ_i , and pass/fail table.

F.2 Reproducible worked example and artifacts

Notebook and inputs. A single public notebook reproduces the map $s \mapsto (\theta_E, \Delta t, h)$ for a toy SIS lens and a synthetic siren. Frozen inputs: $\theta_E^{(0)} = 1.20 \text{ arcsec}$, $\Delta t^{(0)} = 10.00 \text{ days}$, $h^{(0)} = 1.00 \times 10^{-21}$ at $f = 100 \text{ Hz}$, injection $s_{\text{inj}} = 0.95$, random seed SEED=20250301.

Deterministic outputs (written by the notebook).

- artifacts/F1_values.png — value panel $(\theta_E, \Delta t, h)$,
- artifacts/F2_quad.png — plot of $R_{\Delta t}$ vs. R_θ^2 ,
- artifacts/F3Bars.png — bars of (s_{T1}, s_{T2}, s_{T3}) ,
- artifacts/checksums.txt — SHA256 of the three images,
- artifacts/env.txt — Python/platform and package versions,
- artifacts/code_hash.txt — repository or cell-bundle hash (registration record).

Registration statement. Each release includes (i) a Git commit hash of the analysis repository; (ii) a manifest of data object IDs/versions; (iii) the notebook’s environment capture. The artifact files above constitute the reproducibility record.